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EC / EE / ME / CE

ENGINEERING MATHEMATICS

MODULE – 2

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Chapter - 1 : Differential Equations

1.1 Introduction

Many practical problems in Mechanical and Electrical Engineering are formulated by finding how one quantity is related to, or depends upon another quantity. Differential equations are used to represent this relationship.

Derivatives are simply interpreted as rates.

For Example :

If x is a function of t then $\frac{dx}{dt}$ is the rate of change of x with respect to t.

Derivatives of higher order represent rate of rates.

For Example :

If x denotes the displacement of a particle then $\frac{d^2x}{dt^2}$ represents the acceleration.

A **differential equation** is defined as an equation containing derivatives of various orders and the variables.

For Example :

$$\frac{d^3y}{dx^3} + 10x\frac{dy}{dx} = 6\cos x$$

$$\frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} = 20$$

$$\frac{dx}{dt} + \frac{dy}{dt} = 9$$

are all differential equations.

Types of Differential Equations

Differential equations are classified as

- Ordinary Differential Equations
- Partial Differential Equations

Ordinary Differential Equations

Differential equations which involve one independent variable are called ordinary differential equation.

Partial Differential Equations

If the differential equation involves more than one independent variable with respect to them, then it is called a partial differential equation.

Ordinary Differential Equations

1.2 Order and Degree of a Differential Equation

Order

The order of a differential equation is the order of the highest order derivative occurring in the equation.

For Example :

$\frac{d^3y}{dx^3} + 8x\frac{dy}{dx} = 9\cos x$ is of order 3

But the equation

$$4\frac{d^3y}{dx^3} + 5\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + \int y dx = \frac{x^2}{9}$$ is not of order 3 (though it appears)

To find its order eliminate $\int y dx$ so differentiating with respect to we get

$$4\frac{d^4y}{dx^4} + 5\frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} + y = \frac{2x}{9}$$

Therefore, its order is 4.

Note : **The order of the equation**

$$\frac{d^2y}{dx^2} + 9y = 2\int x dx \text{ is 2 because } \int x dx = \frac{x^2}{2}$$

Here the integral involves a variable same as the variable of integration.

Degree

The degree of a differential equation is the degree (or power) of the highest order derivative occurring in the equation after the equation has been made rational and integral in all of its derivatives.

Note : $y = x\frac{dy}{dx} + \cos\frac{dy}{dx}$

Here degree cannot be determined because the differential equation cannot be written as a rational integral form.

Solved Example 1 :

Find the order and degree of following :

(i) $y' = 2x^2$

Solution :

Order = 1

Degree = 1

(ii) $\ddot{y} + 3\dot{y} + 2y = 4x$

Solution :

Order = 2

Degree = 1

(iii) $x^2y'' - xy' + 9y = \log x$

Solution :

Order = 2

Degree = 1

(iv) $y'\,y'' + y^3 = x^3$

Solution :

Order = 2

Degree = 1

(v) $[1 + (y')^3]^{1/2} = 10y$

Solution :

Order = 1

Degree = 3

(vi) $y' = \sqrt{x} + \sqrt{y}$

Solution :

Order = 1

Degree = 1

(vii) $(y''')^2 = 1 + (y')^3$

Solution :

Order = 3

Degree = 2

(viii) $y''' + 8y'' + 2y' + y = 0$

Solution :

Order = 3

Degree = 1

(ix) $\sigma = \dfrac{\left\{1 + \left(\dfrac{dy}{dx}\right)^3\right\}^{3/2}}{\dfrac{d^3y}{dx^3}}$

Solution :

$$\sigma^2\left(\frac{d^3y}{dx^3}\right)^2 = \left\{1 + \left(\frac{dy}{dx}\right)^3\right\}^3$$

Order = 3

Degree = 2

(x) $\dfrac{d}{dx}\left\{x^2\left(\dfrac{d^2y}{dx^2}\right)^3\right\} + \cos(xy) = e^x$

Solution :

$$3\left(\frac{d^2y}{dx^2}\right)^2\frac{d^3y}{dx^3}(x^2) + \left(\frac{d^2y}{dx^2}\right)^3(2x) + \cos(xy) = e^x$$

Order = 3

Degree = 1

(xi) $y = x\dfrac{dy}{dx} + \dfrac{10}{\dfrac{dy}{dx}}$

Solution :

$$y\frac{dy}{dx} = x\left(\frac{dy}{dx}\right)^2 + 10$$

Order = 1

Degree = 2

(xii) $y = x\dfrac{d^2y}{dx^2} + 15\sqrt{1+\left(\dfrac{d^2y}{dx^2}\right)^2}$

Solution :

$$\left(y - x\frac{d^2y}{dx^2}\right)^2 = 225\left[1+\left(\frac{d^2y}{dx^2}\right)^2\right]$$

$$y^2 - 2xy\frac{d^2y}{dx^2} + x^2\left(\frac{d^2y}{dx^2}\right)^2 = 225\left[1+\left(\frac{d^2y}{dx^2}\right)^2\right]$$

Order = 2

Degree = 2

1.3 Formation of Differential Equations

Solved Example 2 :

Obtain the differential equation whose general solution is

$y = A\cos 2x + B\sin 2x$

Solution :

The equation has two arbitrary constants hence differentiating successively two times we get

$y' = -2A\sin 2x + 2B\cos 2x$

$y'' = -4A\cos 2x - 4B\sin 2x$

$= -4y$

or $y'' + 4y = 0$

This is linear second order differential equation.

Solved Example 3 :

Find the differential equation whose G.S. is

$y = bx + \dfrac{1}{b} \qquad b \neq 0$

Solution :

This equation has one arbitrary constant hence differentiating once, we get

$y' = b$

eliminating b we obtain

$y = xy' + \dfrac{1}{y'}$

or $x(y')^2 - yy' + 1 = 0$

This is first order, second degree non linear differential equation.

Solved Example 4 :

Find the differential equation of all straight lines passing through the origin.

Solution :

Straight line equation passing through origin is

$y = mx$, where m is a constant

Differential w.r.t. x we get

$$\frac{dy}{dx} = m$$

Eliminating (m) we get

$$\frac{dy}{dx} = \frac{y}{x}$$

Hence $y = x\dfrac{dy}{dx}$ is the differential equation.

Solved Example 5 :

Find the differential equation of the family of circle of fixed radius r with centres on y axis.

Solution :

Equation of the family of circles of fixed radius with centres on y axis is

$$x^2 + (y - a)^2 = r^2 \qquad \ldots(1)$$

Differentiating w.r.t. x we get,

$$2x + 2(y - a)\, y' = 0 \qquad \ldots(2)$$

$$(y - a) = -\frac{x}{y'}$$

eliminating (y – a) from equation (1)

$$x^2 + \frac{x^2}{(y')^2} = r^2$$

$$\therefore \quad (x^2 - r^2)(y')^2 + x^2 = 0$$

This is the differential equation.

Solved Example 6 :

Find the differential equation of the family of circles touching Y–axis at the origin and centres on X–axis.

Solution :

Equation of the family of circles of fixed radius r with centres on x–axis is

$$(x - r)^2 + y^2 = r^2$$

i.e. $x^2 + y^2 - 2xr = 0 \qquad \ldots(1)$

Differentiating w.r.t. x we have

$$2x + 2y\frac{dy}{dx} - 2r = 0$$

or $\quad r = x + y\dfrac{dy}{dx}$

Substituting in (1) we get

$$x^2 + y^2 - 2x\left(x + y\frac{dy}{dx}\right) = 0$$

$$\therefore \quad 2xy\frac{dy}{dx} + x^2 - y^2 = 0$$

This is the required differential equation.

1.4 First Order First Degree Differential Equations

General form of the equation is

$$M + N\frac{dy}{dx} = 0 \qquad \text{or} \qquad Mdx + Ndy = 0$$

where M and N are functions of x and y.

Variable Seperable Form

General form : $f(x)\, dx = \phi(y)\, dy$

General solution : $\int f(x)\,dx = \int \phi(y)\,dy + C$ …C is arbitrary constant

Solved Example 7 :

Solve $x(4y-3)dx+(x^2+1)dy=0$

Solution :

Dividing by $(4y-3)(x^2+1)$ we get

$$\frac{x}{x^2+1}dx+\frac{1}{4y-3}dy=0$$

Integrating

$$\int\frac{x}{x^2+1}dx+\int\frac{dy}{4y-3}=C$$

$$\frac{1}{2}\log(x^2+1)+\frac{1}{4}\log(4y-3)=C$$

$$\ldots\int\frac{f'(x)}{f(x)}=\log f(x)$$

or $\frac{1}{2}\log(x^2+1)+\frac{1}{4}\log(4y-3)=\log C_1$

$$\ldots C=\log C_1$$

$$\log(x^2+1)+\frac{1}{2}\log(4y-3)=2\log C_1$$

$$(x^2+1)\sqrt{4y-3}=C_1^2$$

Solved Example 8 :

Solve $(8+x)\frac{dy}{dx}+1=5e^{-y}$

Solution :

$$(8+x)dy+dx=5e^{-y}dx$$

$$(8+x)dy=(5e^{-y}-1)dx$$

$$\frac{dx}{x+8}=\frac{e^y dy}{5-e^y}$$

$$\frac{dx}{x+8}+\frac{e^y}{e^y-5}dy=0$$

$$\log(x+8)+\log(e^y-5)=\log C$$

$$(x+8)(e^y-5)=C$$

Solved Example 9 :

Solve $y' - 2y + a = 0$, a is a constant

Solution :

$$\frac{dy}{dx}=2y-a$$

$$\therefore\quad \frac{dy}{2y-a}=dx$$

$$\int\frac{dy}{2y-a}=\int dx$$

$$\frac{1}{2}\log(2y-a)=x+C_1$$

$$2y-a=e^{2x}\cdot e^{2C_1}$$

or $2y-a=ke^{2x}$ $\qquad k=e^{2C_1}$

Solved Example 10 :

Find the curve through the point (1, 0) and having at each of its points the slope –x/y

Solution :

Slope = $\frac{dy}{dx}=-\frac{x}{y}$

Separating the variables

$\therefore$ y dy = –x dx

Integrating

$$\frac{y^2}{2}=-\frac{x^2}{2}+C \qquad \ldots(1)$$

$$y^2=-x^2+2C$$

y(1) = 0 i.e. $y\to 0$, $x\to 1$

0 = –1 + 2C $\quad\therefore\quad$ C = 1/2

Substituting C in (1)

$$\frac{y^2}{2} = -\frac{x^2}{2} + \frac{1}{2} \quad \therefore \quad x^2 + y^2 = 1$$

Solution curves are circles.

Solved Example 11 :

$$\frac{dy}{dx} = (x + 4y + 3)^2$$

Solution :

Put $\quad v = x + 4y + 3$

$$\frac{dv}{dx} = 1 + 4\frac{dy}{dx} \quad \text{or} \quad \frac{dy}{dx} = \frac{1}{4}\left(\frac{dv}{dx} - 1\right)$$

Substituting in equation $\quad \frac{1}{4}\left(\frac{dv}{dx} - 1\right) = v^2$

$$\therefore \quad \frac{dv}{dx} - 1 = 4v^2 \qquad \therefore \quad \frac{dv}{dx} = 4v^2 + 1$$

$$\therefore \quad \frac{dv}{4v^2 + 1} = dx \quad \text{separating the variables}$$

$$\text{or} \quad \frac{dv}{(2v)^2 + 1} = dx$$

Integrating $\quad \int \frac{dv}{(2v)^2 + 1} = \int dx$

$$\frac{1}{2}\tan^{-1}(2v) = x + C$$

$$\text{or} \quad \frac{1}{2}\tan^{-1} 2(x + 4y + 3) = x + C$$

Solved Example 12 :

Solve $\quad \frac{dy}{dx} = \sin(x + y)$

Solution :

Put $\quad x + y = v$

$$\frac{dy}{dx} + 1 = \frac{dv}{dx}$$

$$\therefore \quad \frac{dy}{dx} = \frac{dv}{dx} - 1$$

Substituting back in equation

$$\frac{dv}{dx} - 1 = \sin v$$

$$\frac{dv}{dx} = 1 + \sin v$$

Separating variables and integrating

$$\int \frac{dv}{1 + \sin v} = \int dx$$

Put $\theta = \tan\frac{v}{2}$

$$\therefore \quad dv = \frac{2d\theta}{1 + \theta^2}$$

$$\sin v = \frac{2\theta}{1 + \theta^2}$$

$$\therefore \quad \int \frac{\frac{2d\theta}{1 + \theta^2}}{1 + \frac{2\theta}{1 + \theta^2}} = x + C$$

$$\therefore \quad \int \frac{2d\theta}{1 + 2\theta + \theta^2} = x + C$$

$$\therefore \quad 2\int \frac{d\theta}{(\theta + 1)^2} = x + C$$

$$\therefore \quad -\frac{2}{\theta + 1} = x + C$$

$$\therefore \quad \frac{-2}{\tan\frac{x + y}{2} + 1} = x + C$$

$$\therefore \quad \tan\frac{x + y}{2} + 1 = \frac{-2}{x + C}$$

$$\text{or} \quad \tan\frac{x + y}{2} + 1 + \frac{2}{x + C} = 0$$

Homogenous Equations

A differential equation Mdx + Ndy = 0 is homogenous if M and N are homogenous in x and y.

Key substitution : y = vx

Solved Example 13 :

$$\frac{dy}{dx} = \frac{y}{x} + \sin\frac{y}{x}$$

Solution :

Put y = vx

$$\therefore \quad \frac{dy}{dx} = x\frac{dv}{dx} + v$$

$$v + x\frac{dv}{dx} = v + \sin v$$

$$\therefore \quad x\frac{dv}{dx} = \sin v$$

Separating variables and integrating

$$\int \frac{dv}{\sin v} = \int \frac{dx}{x}$$

Put $\theta = \tan\frac{v}{2}$

$$\therefore \quad dv = \frac{2d\theta}{1+\theta^2}$$

$$\sin v = \frac{2\theta}{1+\theta^2}$$

$$\therefore \quad \int \frac{\frac{2d\theta}{1+\theta^2}}{\frac{2\theta}{1+\theta^2}} = \int \frac{dx}{x}$$

$$\int \frac{d\theta}{\theta} = \int \frac{dx}{x}$$

$$\log\left(\tan\frac{v}{2}\right) = \log x + C$$

$$\log\left(\tan\frac{y}{2x}\right) = \log x + \log C_1$$

$$C = \log C_1$$

$$\tan\frac{y}{2x} = xC_1$$

Solved Example 14 :

Solve $x(y-x)\frac{dy}{dx} = y(y+x)$

Solution :

The given equation is

$$\frac{dy}{dx} = \frac{y(y+x)}{x(y-x)}$$

Put y = vx

then $\frac{dy}{dx} = v + x\frac{dv}{dx}$

$$\therefore \quad v + x\frac{dv}{dx} = \frac{vx(vx+x)}{x(vx-x)}$$

$$\therefore \quad v + x\frac{dv}{dx} = \frac{v(v+1)}{v-1}$$

$$\therefore \quad x\frac{dv}{dx} = \frac{v(v+1)}{v-1} - v$$

$$\therefore \quad x\frac{dv}{dx} = \frac{v(v+1) - v(v-1)}{v-1} = \frac{2v}{v-1}$$

$$\frac{(v-1)dv}{2v} = \frac{dx}{x} \quad \text{separating variables}$$

Integrating

$$\int\left(\frac{1}{2}-\frac{1}{2v}\right)dv=\int\frac{dx}{x}$$

$$\therefore\quad \frac{v}{2}-\frac{1}{2}\log v=\log x+C$$

$$\therefore\quad \frac{y}{2x}-\frac{1}{2}\log\frac{y}{x}=\log x+C$$

$$\therefore\quad \frac{y}{x}-\log\frac{y}{x}=2\log x+2C$$

$$\therefore\quad \frac{y}{x}-\log\frac{y}{x}-\log x^2=C_1$$

$C_1 = 2C$

$$\therefore\quad \frac{y}{x}-\log yx=C_1$$

Solved Example 15 :

Solve $\left(x^2+4y^2+xy\right)dx-x^2dy=0$

Solution :

The given equation is

$$\frac{dy}{dx}=\frac{x^2+4y^2+xy}{x^2}$$

Substituting y = vx

$$\frac{dy}{dx}=v+x\frac{dv}{dx}$$

$$\therefore\quad v+x\frac{dv}{dx}=\frac{x^2+4v^2x^2+xvx}{x^2}$$

$$\therefore\quad x\frac{dv}{dx}=4v^2+1$$

$$\therefore\quad \frac{dv}{1+4v^2}=\frac{dx}{x}$$

$$\therefore\quad \frac{1}{2}\tan^{-1}\frac{2v}{1}=\log x+C$$

$$\therefore\quad \frac{1}{2}\tan^{-1}\frac{2y}{x}=\log x+\log C_1$$

$$\ldots C=\log C_1$$

Solved Example 16 :

Solve $\dfrac{dy}{dx}=\dfrac{y^3+3x^2y}{x^3+3xy^2}$

Solution :

Put y = vx

$$\therefore\quad \frac{dy}{dx}=v+x\frac{dv}{dx}$$

$$\therefore\quad v+x\frac{dv}{dx}=\frac{v^3+3v}{1+3v^2}$$

$$\therefore\quad x\frac{dv}{dx}=\frac{v^3+3v}{1+3v^2}-v=\frac{2v\left(1-v^2\right)}{1+3v^2}$$

$$\therefore\quad \frac{1+3v^2}{v\left(1-v^2\right)}dv=2\frac{dx}{x}$$

$$\therefore\quad 2\frac{dx}{x}=\frac{1}{v}-\frac{2}{1+v}+\frac{2}{1-v}$$

(By partial fraction)

$$\therefore\quad 2\log x=\log v-2\log(1+v)$$

$$-2\log(1-v)+\log C$$

$$\therefore\quad x^2=\frac{Cv}{(1+v)^2(1-v)^2}$$

$$\therefore\quad x^2(1+v)^2(1-v)^2=Cv$$

$$\therefore\quad x^2\left(1-v^2\right)^2=Cv$$

$$\therefore\quad x^2\left(1-\frac{y^2}{x^2}\right)^2=C\left(\frac{y}{x}\right)$$

$$\therefore\quad \left(x^2-y^2\right)^2=Cxy$$

Solved Example 17 :

Solve : $\left(3xy+y^2\right)dx+\left(x^2+xy\right)dy=0$

$y(1) = 1$

Solution :

The equation is

$$\frac{dy}{dx}=-\frac{\left(3xy+y^2\right)}{x^2+xy}$$

Put y = vx

$$\frac{dy}{dx}=v+x\frac{dv}{dx}$$

$$\therefore \quad v+x\frac{dv}{dx}=-\frac{\left(3x(vx)+v^2x^2\right)}{x^2+x(vx)}$$

$$\therefore \quad x\frac{dv}{dx}=\frac{-3v-v^2}{1+v}-v$$

$$\therefore \quad x\frac{dv}{dx}=\frac{-3v-v^2-v-v^2}{1+v}$$

$$\therefore \quad x\frac{dv}{dx}=\frac{-4v-2v^2}{1+v}$$

$$\therefore \quad x\frac{dv}{dx}=\frac{-2v(2+v)}{1+v}$$

Separating variables

$$\frac{1+v}{v(2+v)}dv=-\frac{2}{x}dx$$

Simplifying $\dfrac{1+v}{v(2+v)}=\dfrac{A}{v}+\dfrac{B}{2+v}$

$1 + v = A(2 + v) + B(v)$

when $v = -2$ $\quad B = 1/2$

when $v = 0$ $\quad A = 1/2$

$$\therefore \quad \frac{1+v}{v(2+v)}=\frac{1}{2}\left[\frac{1}{v}+\frac{1}{2+v}\right]$$

$$\therefore \quad \frac{1}{2}\left[\frac{1}{v}+\frac{1}{2+v}\right]dv=-\frac{2dx}{x}$$

Integrating we get

$$\frac{1}{2}\log v+\frac{1}{2}\log(2+v)=-2\log x+\log C$$

$$\log v+\log(2+v)=-4\log x+2\log C$$

$$\log\left[v(2+v)\right]+4\log x=2\log C$$

$$v(2+v)x^4=C^2$$

$$\frac{y}{x}\left(2+\frac{y}{x}\right)x^4=C^2$$

$$y(2x+y)x^2=C^2$$

when $y = 1, x = 1$

$3 = C^2$

The particular solution is

$$y(2x+y)x^2=3$$

Solved Example 18 :

Solve $x^2ydx-\left(x^3+y^3\right)dy=0$

Solution :

The equation is

$$\frac{dy}{dx}=\frac{x^2y}{x^3+y^3}$$

Put $y = vx$

$$\frac{dy}{dx}=v+x\frac{dv}{dx}$$

$$v+x\frac{dv}{dx}=\frac{x^2vx}{x^3+v^3x^3}=\frac{v}{1+v^3}$$

$$\therefore \quad x\frac{dv}{dx}=\frac{v}{1+v^3}-v=\frac{-v^4}{1+v^3}$$

$$\therefore \quad \frac{\left(1+v^3\right)}{v^4}dv=-\frac{dx}{x}$$

$$\therefore \quad \frac{1}{v^4}dv+\frac{1}{v}dv=-\frac{dx}{x}$$

Integrating we get,

$$-\frac{1}{3}\frac{1}{v^3}+\log v=-\log x+\log C$$

$$\log v+\log x-\log C=\frac{1}{3v^3}$$

$$\frac{vx}{C}=e^{1/3v^3}$$

$$\frac{y}{C}=e^{\frac{1}{3\left(\frac{y}{x}\right)^3}}$$

$$\therefore \quad y=Ce^{\frac{1}{3\left(\frac{y}{x}\right)^3}}$$

Non Homogenous Equations

General Form : $\dfrac{dy}{dx}=\dfrac{a_1x+b_1y+c_1}{a_2x+b_2y+c_2}$

Case 1 : $\dfrac{a_1}{a_2}=\dfrac{b_1}{b_2}$

Case 2 : $\dfrac{a_1}{a_2}\neq\dfrac{b_1}{b_2}$

In this case, the equation is reduced to homogenous form.

Solved Example 19 :

Based on Case 1

Solve : $(2x+y+1)dx+(4x+2y-1)dy=0$

Solution :

The equation is

$$\frac{dy}{dx}=\frac{-(2x+y+1)}{4x+2y-1}$$

$$=\frac{-(2x+y+1)}{2(2x+y)-1} \qquad \left\{\frac{a_1}{a_2}=\frac{b_1}{b_2}\right\}$$

Put 2x + y = u

$$\therefore \quad 2+\frac{dy}{dx}=\frac{du}{dx}$$

$$\therefore \quad \frac{dy}{dx}=\frac{du}{dx}-2$$

$$\therefore \quad \frac{du}{dx}-2=\frac{-(u+1)}{2u-1}$$

$$\therefore \quad \frac{du}{dx}=\frac{-u-1+4u-2}{2u-1}$$

$$\therefore \quad \frac{du}{dx}=\frac{3u-3}{2u-1}=\frac{3(u-1)}{2u-1}$$

$$\therefore \quad \frac{(2u-1)}{3u-3}du=dx$$

Simplifying $\dfrac{2u-1}{3u-3}=A+B\dfrac{1}{3u-3}$

$$\therefore \quad 2u-1=A(3u-3)+B$$

Comparing coefficients of u

$\therefore$ 2u – 3A $\quad\therefore$ A = 2/3

Substituting A in above equation

$$2u-1=\frac{2}{3}(3u-3)+B \qquad \therefore \quad B=1$$

Now $\displaystyle\int\left(\frac{2}{3}+\frac{1}{3u-3}\right)du=\int dx$

Integrating

$$\therefore \quad \frac{2}{3}u + \frac{1}{3}\log(u-1) = x + C$$

$$\therefore \quad 2(2x+y) + \log(2x+y-1) = 3x + C_1$$

$$\ldots C_1 = 3C$$

$$x + 2y + \log(2x+y-1) = C_1$$

Solved Example 20 :

Based on Case 2

Solve $\dfrac{dy}{dx} = \dfrac{x+2y-3}{2x+y-3}$

Solution :

$\dfrac{1}{2} \neq \dfrac{2}{1}$ $\qquad \therefore \dfrac{a_1}{a_2} \neq \dfrac{b_1}{b_2}$

Let x = X + h

y = Y + k $\qquad$ h, k are constants

$$\frac{dy}{dx} = \frac{dY}{dX}$$

$$\therefore \quad \frac{dY}{dX} = \frac{X+h+2(Y+k)-3}{2(X+h)+(Y+k)-3}$$

$$= \frac{X+h+2Y+2k-3}{2X+2h+Y+k-3}$$

$$= \frac{X+2Y+h+2k-3}{2X+Y+2h+k-3} \qquad \ldots(1)$$

Solving h + 2k – 3 = 0

& 2h + k – 3 = 0

we get h = 1, k = 1

Substituting in equation (1)

$$\frac{dY}{dX} = \frac{X+2Y}{2X+Y}$$

This is homogenous equation in X, Y

Put Y = VX we get

$$V + X\frac{dV}{dX} = \frac{X+2VX}{2X+VX}$$

$$\therefore \quad X\frac{dV}{dX} = \frac{1+2V}{2+V} - V$$

$$\therefore \quad X\frac{dV}{dX} = \frac{1+2V-2V-V^2}{2+V}$$

$$\therefore \quad X\frac{dV}{dX} = \frac{1-V^2}{2+V}$$

$$\therefore \quad \frac{(2+V)}{1-V^2}dV = \frac{dX}{X}$$

Simplifying , $\dfrac{2+V}{1-V^2} = \dfrac{A}{1-V} + \dfrac{B}{1+V}$

$$2 + V = A(1+V) + B(1-V)$$

when $\quad V = -1 \qquad B = 1/2$

$\qquad\quad V = 1 \qquad A = 3/2$

$$\int \frac{3}{2}\left(\frac{1}{1-V}\right)dV + \int \frac{1}{2}\left(\frac{1}{1+V}\right)dV = \int \frac{dX}{X}$$

$$\therefore \quad -\frac{3}{2}\log(-V+1) + \frac{1}{2}\log(V+1) = \log X + C$$

$$\therefore \quad -3\log(-V+1) + \log(V+1)$$

$$= 2\log X + 2\log C_1 \qquad C = \log C_1$$

$$\therefore \quad (-V+1)^{-3}\cdot(V+1) = X^2 \cdot C_2 \qquad C_2 = C_1^2$$

$$\therefore \quad \left(-\frac{Y}{X}+1\right)^{-3}\left(\frac{Y}{X}+1\right) = X^2 \cdot C_2$$

$$\therefore \quad \left(\frac{X-Y}{X}\right)^{-3}\left(\frac{Y+X}{X}\right) = X^2 C_2$$

$$\therefore \quad Y + X = (X-Y)^3 C_2$$

X = x – h = x – 1 and Y = y – k = y – 1

$y - 1 + x - 1 = (x - 1 - y + 1)^3 C_2$

$$\therefore \quad (x+y-2) = (x-y)^3 C_2$$

1.5 Exact Differential Equations

The equation Mdx + Ndy = 0 is said to be an exact differential equation if and only if there exists a function f(x, y) such that

$$\frac{\partial f}{\partial x} = M(x,y) \text{ and } \frac{\partial f}{\partial y} = N(x,y)$$

Condition for a Differential Equation to be exact :

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

General Solution :

$$\int_{\substack{\text{Keeping y}\\ \text{constant}}} \mathbf{Mdx} + \int [\textbf{Terms in N which do not involve x}]\mathbf{dy} = \mathbf{C}$$

Solved Example 21 :

Solve : $(1+e^x)dx + ydy = 0$

Solution :

Here $M = 1 + e^x$

$N = y$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$0 = 0$

Equation is exact.

General solution is

$$\int_{\text{Y const}} (1+e^x)dx + \int ydy = C$$

$$\therefore \quad x + e^x + \frac{y^2}{2} = C$$

Solved Example 22 :

Solve :

$$(x^2 - 4xy - 2y^2)dx + (y^2 - 4xy - 2x^2)dy = 0$$

Solution :

Here $M = x^2 - 4xy - 2y^2$

$N = y^2 - 4xy - 2x^2$

$$\frac{\partial M}{\partial y} = -4x - 4y$$

$$\frac{\partial N}{\partial x} = -4x - 4y$$

Hence the equation is exact.

General solution is

$$\int_{\text{y const}} (x^2 - 4xy - 2y^2) + \int y^2 dy = C$$

$$\frac{1}{3}x^3 - 2x^2y - 2y^2x + \frac{1}{3}y^3 = C$$

or $\quad x^3 + y^3 - 6xy(x+y) = C_1 \qquad C_1 = 3C$

Solved Example 23 :

Solve : $xdx + ydy - \left(\frac{xdy - ydx}{x^2 + y^2}\right) = 0$

Solution :

Here $M = x + \frac{y}{x^2 + y^2} \qquad N = y - \frac{x}{x^2 + y^2}$

$$\therefore \quad \frac{\partial M}{\partial y} = \frac{x^2 - y^2}{(x^2 + y^2)^2} \qquad \frac{\partial N}{\partial y} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$$

The equation is exact.

General solution is

$$\int_{y\,const}\left(x+\frac{y}{x^2+y^2}\right)dx+\int ydy=C$$

$$\therefore\quad \frac{1}{2}x^2+\tan^{-1}\left(\frac{x}{y}\right)+\frac{y^2}{2}=C$$

$$\therefore\quad x^2+y^2+2\tan^{-1}\left(\frac{x}{y}\right)=C_1 \qquad C_1=2C$$

Solved Example 24 :

Solve : $\sinh x\cos ydx-\cosh x\sin ydy=0$

Solution :

Here M = sin hx cos y

N = – cosh x sin y

Equation is exact.

General solution is

$$\int_{y\,const}\sinh x\cos ydx+\int 0dy=C$$

cos y cos hx = C

Solved Example 25 :

Solve :

$$\left(2x^2y+4x^3-12xy^2+3y^2-xe^y+e^{2x}\right)dy+$$

$$\left(12x^2y+2xy^2+4x^3-4y^3+2ye^{2x}-e^y\right)dx=0$$

Solution :

Here

M = $(12x^2y+2xy^2+4x^3-4y^3+2ye^{2x}-e^y)$

N = $2x^2y+4x^3-12xy^2+3y^2-xe^y+e^2x$

$$\frac{\partial M}{\partial y}=12x^2+4xy-12y^2+2e^{2x}-e^y$$

$$\frac{\partial N}{\partial x}=4xy+12x^2-12y^2-e^y+2e^{2x}$$

$$\frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}$$

Equation is exact.

General solution is

$$\int_{y\,const}\left(12x^2y+2xy^2+4x^3-4y^3+2ye^{2x}-e^y\right)dx$$

$$+\int 3y^2dy=C$$

$$12\frac{x^3}{3}y+\frac{2x^2}{2}y^2+\frac{4x^4}{4}-4y^3x$$

$$+2y\frac{e^{2x}}{2}-e^y\cdot x+y^3=C$$

$$4x^3y+x^2y^2+x^4-4y^3x+ye^{2x}-e^yx+y^3=C$$

Solved Example 26 :

Determine for what values of a and b the differential equation is exact and obtain the differential equation.

$$\left(y+x^3\right)dx+\left(ax+by^3\right)dy=0$$

Solution :

Here

M = Y + x^3 N = ax + by^3

$$\frac{\partial M}{\partial y}=1 \qquad \frac{\partial N}{\partial x}=a$$

Hence if a = 1 equation is exact irrespective of value of b.

General solution is

$$\int_{y\,const}\left(y+x^3\right)dx+\int by^3dy=C$$

$$xy+\frac{x^4}{4}+\frac{by^4}{4}=C$$

1.6 Differential Equations reducible to Exact form

Equations which are not exact can be converted to exact differential by multiplying by some factor known as Integrating factor.

For Example :

$3ydx - 2xdy = 0$

Here equation is not exact

Multiplying by $\frac{1}{xy}$ we get

$$\frac{3}{x}dx - \frac{2}{y}dy = 0$$

Now equation is exact.

Rules for finding Integrating Factor

- By Inspection :

Terms	I. F.	Differential
xdy – ydx	$\frac{1}{x^2}$	$\frac{xdy - ydx}{x^2} = d\left(\frac{y}{x}\right)$
	$\frac{1}{xy}$	$\frac{dy}{y} - \frac{dx}{x} = d\log\left(\frac{y}{x}\right)$
	$\frac{1}{y^2}$	$\frac{xdy - ydx}{y^2} = d\left(\frac{-x}{y}\right)$
	$\frac{1}{x^2 + y^2}$	$\frac{xdy - ydx}{x^2 + y^2} = d\left(\tan^{-1}\frac{y}{x}\right) = -d\tan^{-1}\frac{x}{y}$
xdy + ydx	$\frac{1}{xy}$	$\frac{xdy + ydx}{xy} = d\log(xy)$
	$\frac{1}{x^2 + y^2}$	$\frac{xdx + ydy}{x^2 + y^2} = \frac{1}{2}\left[\log\left(x^2 + y^2\right)\right]$

- If Mdx + Ndy = 0 is homogenous then

$$I.F. = \frac{1}{Mx + Ny}$$

- If $Mdx + Ndy = 0$

  and $M = y\, f_1(xy)$, $N = x\, f_2(xy)$ then

$$\text{I.F.} = \frac{1}{Mx - Ny}$$

- If Mdx + Ndy = 0s

  and $\dfrac{\dfrac{\partial M}{\partial y} - \dfrac{\partial N}{\partial x}}{N} = f(x)$ function of x alone or constant

$$\text{I.F.} = e^{\int f(x)\, dx}$$

- If Mdx + Ndy = 0

  and $\dfrac{\dfrac{\partial N}{\partial x} - \dfrac{\partial M}{\partial y}}{M} = \phi(y)$ a function of y along or constant.

$$\text{I.F.} = e^{\int \phi(y)\, dy}$$

Solved Example 27 :

Solve : $(y-1)dx - xdy = 0$

Solution :

Here M = y – 1 N = –x

$\dfrac{\partial M}{\partial y} = 1$; $\dfrac{\partial N}{\partial x} = -1$

$\dfrac{\partial M}{\partial y} \neq \dfrac{\partial N}{\partial x}$ $\therefore$ not exact

$$\frac{\dfrac{\partial M}{\partial y} - \dfrac{\partial N}{\partial x}}{N} = -\frac{2}{x} = f(x) \quad [\text{Rule 4}]$$

$$\therefore \quad \text{I.F.} = \int_e -\frac{2}{x}dx = e^{-2\log x} = e^{\log x^{-2}} = \frac{1}{x^2}$$

Multiplying equation with I. F. we get

$$\frac{(y-1)}{x^2}dx - \frac{x}{x^2}dy = 0$$

or $$\frac{(y-1)}{x^2}dx - \frac{1}{x}dy = 0$$

General solution is

$$\int_{y\,\text{const}} \frac{y-1}{x^2}dx + \int 0dy = C$$

$$\int \frac{y}{x^2}dx - \int \frac{1}{x^2}dy = C$$

$$\therefore \quad \frac{-y}{x} + \frac{1}{x} = C$$

$\therefore$ 1 – y = xC

Solved Example 28 :

Solve :

$$(3x^2y^3e^y + y^3 + y^2)dx + (x^3y^3e^y - xy)dy = 0$$

Solution :

Here $M = 3x^2y^3e^y + y^3 + y^2$

$N = x^3y^3e^y - xy$

$$\frac{\partial M}{\partial y} = 9x^2y^2e^y + 3y^3x^2e^y + 3y^2 + 2y$$

$$\frac{\partial N}{\partial x} = 3x^2y^3e^y - y$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} \quad \text{not exact.}$$

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = -\frac{\left(9x^2y^2e^y + 3y^2 + 3y\right)}{y\left(3x^2y^2e^y + y^2 + y\right)} = -\frac{3}{y}$$

$$\text{I.F.} = \int_e \frac{-3}{y}dy = e^{-3\log y} = y^{-3} = \frac{1}{y^3}$$

Multiplying equation with I.F.

$$\left(\frac{3x^2e^yy^3 + y^3 + y^2}{y^3}\right)dx + \left(\frac{x^3y^3e^y - xy}{y^3}\right)dy = 0$$

$$\therefore \quad \left(3x^2e^y + 1 + \frac{1}{y}\right)dx + \left(x^3e^y - \frac{x}{y^2}\right)dy = 0$$

General solution is

$$\int_{y\,\text{const}} \left(3x^2e^y + 1 + \frac{1}{y}\right)dx + \int 0\,dy = C$$

$$x^3e^y + x + \frac{x}{y} = C$$

Solved Example 29 :

Solve :

$$\left(3xy^2 - y^3\right)dx + \left(xy^2 - 2x^2y\right)\,dy = 0$$

Solution :

Here $\quad M = 3xy^2 - y^3$

$\quad\quad N = xy^2 - 2x^2y$

The equation is homogenous

$$\text{I.F.} = \frac{1}{Mx + Ny} \qquad \text{[Rule (2)]}$$

$$= \frac{1}{\left(3xy^2 - y^3\right)x + \left(xy^2 - 2x^2y\right)y}$$

$$= \frac{1}{3x^2y^2 - xy^3 + xy^3 - 2x^2y^2} = \frac{1}{x^2y^2}$$

Multiplying equation with I.F.

$$\frac{3xy^2 - y^3}{x^2y^2}dx + \frac{xy^2 - 2x^2y}{x^2y^2}dy = 0$$

$$\left(\frac{3}{x} - \frac{y}{x^2}\right)dx + \left(\frac{1}{x} - \frac{2}{y}\right)dy = 0$$

$$\text{G.S. is} \quad \int_{y\,\text{const}} \left(\frac{3}{x} - \frac{y}{x^2}\right)dx + \int \frac{-2}{y}dy = C$$

$$3\log x + \frac{y}{x} - 2\log y = C$$

$$\log x^3 - \log y^2 + \frac{y}{x} = \log C_1 \qquad C = \log C_1$$

$$\log\frac{x^3}{y^2C_1} = -\frac{y}{x}$$

$$e^{y/x} = \frac{C_1y^2}{x^3}$$

Solved Example 30 :

Solve :

$$(x^2y^2 + 5xy + 2)y\,dx + (x^2y^2 + 4xy + 2)x\,dy = 0$$

Solution :

Here $\quad M = \left(x^2y^2 + 5xy + 2\right)y$

$\quad\quad N = \left(x^2y^2 + 4xy + 2\right)x$

$$\text{I. F.} = \frac{1}{Mx - Ny} \qquad \text{[Rule 3]}$$

$$= \frac{1}{xy\left[x^2y^2 + 5xy + 2 - x^2y^2 - 4xy - 2\right]}$$

$$= \frac{1}{x^2y^2}$$

Multiplying equation with I.F.

$$\frac{(x^2y^2 + 5xy + 2)}{x^2y^2}ydx + \frac{(x^2y^2 + 4xy + 2)x}{x^2y^2}dy = 0$$

$$\left(y + \frac{5}{x} + \frac{2}{x^2y}\right)dx + \left(x + \frac{4}{y} + \frac{2}{xy^2}\right)dy = 0$$

General solution is

$$\int_{y\,\text{const}} \left(y + \frac{5}{x} + \frac{2}{x^2 y} \right) dx + \int \frac{4}{y} dy = C$$

$$yx + 5\log x - \frac{2}{yx} + 4\log y = C$$

$$-\frac{2}{yx} + yx + \log x^5 + \log y^4 = \log C_1$$

$$C = \log C_1$$

$$xy - \frac{2}{xy} = \log\left(\frac{C_1}{x^5 y^4} \right)$$

$$e^{xy - \frac{2}{xy}} = \frac{C_1}{x^5 y^4}$$

Solved Example 31 :

Solve : $x\left(1+y^2\right)dy + y\left(1+x^2\right)dx = 0$

Solution :

The equation is

$$xdy + ydx + xy\left(ydy + xdx\right) = 0$$

equation has terms xdy + ydx and xy

$\therefore$ I.F. $= \dfrac{1}{xy}$ [Rule 1]

Multiplying equation with I.F.

$$\frac{x\,dy + y\,dx}{xy} + y\,dy + x\,dx = 0$$

Integrating,

$$\log(xy) + \frac{y^2}{2} + \frac{x^2}{2} = C$$

Solved Example 32 :

Solve $xdx + ydy + 2\left(x^2 + y^2\right)dx = 0$

Solution :

The equation has terms

xdx + ydy and $x^2 + y^2$

$\therefore$ I.F. $= \dfrac{1}{x^2 + y^2}$

Multiplying equation with I.F.

$$\frac{xdx + ydy}{x^2 + y^2} + 2dx = 0$$

Integrating we get

$$\frac{1}{2}\log\left(x^2 + y^2\right) + 2x = C$$

$$\therefore \quad \log\left(x^2 + y^2\right) = 2C - 4x$$

$$\therefore \quad x^2 + y^2 = C_1 e^{-4x} \qquad C_1 = e^{2C}$$

Solved Example 33 :

Solve : $ydx - xdy + e^{1/x}dx = 0$

Solution :

The equation has terms ydx – xdy and $e^{1/x}$

$\therefore$ I.F. $= \dfrac{1}{x^2}$

Multiplying with I.F.

$$\frac{ydx - xdy}{x^2} + \frac{e^{1/x}}{x^2}dx = 0$$

$$-d\left(\frac{y}{x}\right) - d\left(e^{1/x}\right) = 0$$

Integrating we get

$$\frac{y}{x} + e^{1/x} = C$$

1.7 First Order Linear Differential Equations

The differential equation $\frac{dy}{dx}+Py=Q$ is called a linear differential equation because the dependent variables and its derivatives appear only in the first degree. Here P and Q are functions of x alone or are constants.

Integrating Factor : $e^{\int P\,dx}$

General solution :

$$y\,(I.F.) = \int Q(I.F.)\,dx + C$$

Note : **(1) Linear first order differential equation in x is of form $\frac{dx}{dy}+Px=Q$.**

Here P and Q are functions of Y alone or constant.

Integrating factor : $e^{\int P\,dy}$

General solution : $x(I.F.)=\int Q(I.F.)\,dy+C$

(2) While solving problems on linear equations the coefficient of $\frac{dy}{dx}$ or $\frac{dx}{dy}$ unity.

Solved Example 34 :

Solve $\frac{dy}{dx}+y\cot x=\sin 2x$

Solution :

Here P = cot x Q = sin 2x

I.F. = $e^{\int P dx}=e^{\int \cot x\,dx}=e^{\log \sin x}=\sin x$

General solution is y sinx

$=\int \sin 2x \sin x dx + C$

$=\int \sin 2x \sin x dx$

$=\int 2\sin^2 x \cos x dx$ sin2A = 2 sinA cosA

Put sin x = t

cos x dx = dt

$$\int 2t^2 dt = \frac{2t^3}{3} = \frac{2\sin^3 x}{3}$$

$$y \sin x = \frac{2}{3}\sin^3 x + C$$

or $y = \frac{2}{3}\sin^2 x + C\operatorname{cosec} x$

Solved Example 35 :

Solve : $(x+2y^3)\frac{dy}{dx}=y$

Solution :

The equation is

$$\frac{dx}{dy}=\frac{x+2y^3}{y}$$

$$\therefore \quad \frac{dx}{dy} = \frac{x}{y} + 2y^2$$

Here $P = -\frac{1}{y}$ $\qquad Q = 2y^2$

$$\text{I.F.} = e^{\int P dy} = e^{\int -\frac{1}{y} dy} = e^{-\log y} = y^{-1} = \frac{1}{y}$$

General solution is

$$\frac{x}{y} = \int \frac{2y^2}{y} dy + C$$

$$\therefore \quad \frac{x}{y} = y^2 + C$$

$$\therefore \quad x = y^3 + yC$$

Solved Example 36 :

Solve $x^2 \frac{dy}{dx} + xy = 2x^2 e^{x^2}$

Solution :

The equation is $\frac{dy}{dx} + \frac{y}{x} = 2e^{x^2}$

Here $P = \frac{1}{x}$ $\qquad Q = 2e^{x^2}$

$$\text{I.F.} = e^{\int P dx} = e^{\int \frac{1}{x} dx} = e^{\log x} = x$$

General solution is

$$y\,(\text{I.F.}) = \int Q(\text{I.F.})\,dx + C$$

$$y \cdot x = \int 2e^{x^2} x dx + C = \int e^{x^2} (2x) dx + C$$

Put $x^2 = t$

$2x\,dx = dt$

$$\therefore \quad y \cdot x = \int e^t dt + C$$

$$y \cdot x = e^t + C$$

$$y \cdot x = e^{x^2} + C$$

$$\therefore \quad y = \frac{C + e^{x^2}}{x}$$

Solved Example 37 :

Solve : $\left(1 + y^2\right) dx + \left(x - \tan^{-1} y\right) dy = 0$

Solution :

The equation can be written as

$$\frac{dx}{dy} + \frac{x}{1 + y^2} = \frac{\tan^{-1} y}{1 + y^2}$$

This is linear first order differential equation in x.

Here $P = \frac{1}{1 + y^2}$, $\qquad Q = \frac{\tan^{-1} y}{1 + y^2}$

$$\text{I.F.} = e^{\int P\,dy} = e^{\int \frac{1}{y^2+1} dy} = e^{\tan^{-1} y}$$

General solution is $x(\text{I.F}) = \int Q(\text{I.F})\,dy + C$

$$\therefore \quad x e^{\tan^{-1} y} = \int \left(\frac{e^{\tan^{-1} y}}{1 + y^2} \right) \tan^{-1} y dy$$

Put $t = \tan^{-1} y$ on R.H.S.

$$dt = \frac{1}{1 + y^2} dy$$

$$\therefore \quad x e^{\tan^{-1} y} = \int t e^t dt = e^t (t - 1) + C$$

$$= e^{\tan^{-1} y} \left(\tan^{-1} y - 1\right) + C$$

$$\therefore \quad x = \left(\tan^{-1} y - 1\right) + C e^{-\tan^{-1} y}$$

Solved Example 38 :

Solve $\left(x^2 + 1\right)^3 \frac{dy}{dx} + 4x\left(x^2 + 1\right)^2 y = 1$

Solution :

The equation is

$$\frac{dy}{dx} + \frac{4x\left(x^2 + 1\right)^2}{\left(x^2 + 1\right)^3} y = \frac{1}{\left(x^2 + 1\right)^3}$$

here $P = \dfrac{4x}{x^2+1}$, $Q = \dfrac{1}{\left(x^2+1\right)^3}$

$$\text{I.F.} = e^{\int P\,dx} = e^{\int \frac{4x}{\left(x^2+1\right)}}$$

Put $x^2 = t$, $2x\,dx = dt$

$$\int \frac{2dt}{t+1} = 2\log(t+1) = 2\log\left(x^2+1\right)$$

$$\therefore \quad \text{I.F.} = e^{2\log\left(x^2+1\right)} = \left(x^2+1\right)^2$$

General solution is $Y(\text{I.F}) = \int Q(\text{I.F.})dx + C$

$$\therefore \quad y\left(x^2+1\right)^2 = \int \frac{1}{\left(x^2+1\right)^3}\left(x^2+1\right)^2 dx$$

$$= \int \frac{1}{x^2+1}dx$$

$$\therefore \quad y\left(x^2+1\right)^2 = \tan^{-1}x + C$$

Solved Example 39 :

Solve $x\dfrac{dy}{dx} + (1+2x)y = 1 + xe^{-2x}$

Solution :

$$\frac{dy}{dx} + \frac{(1+2x)}{x}y = \frac{1+xe^{-2x}}{x}$$

Here $P = \dfrac{1+2x}{x}$, $Q = \dfrac{1+xe^{-2x}}{x}$

$$\text{I.F.} = e^{\int P dx} = e^{\int \frac{1+2x}{x}dx}$$

$$\int \frac{1+2x}{x}dx = \int \frac{1}{x}dx + \int 2dx = \log x + 2x$$

$$\therefore \quad \text{I.F.} = e^{\log x + 2x} = x \cdot e^{2x}$$

General solution is $y(\text{I.F.}) = \int Q(\text{I.F})dx + C$

$$\therefore \quad y \cdot x \cdot e^{2x} = \int \frac{1+xe^{-2x}}{x}\left(x \cdot e^{2x}\right)dx$$

$$\therefore \quad y \cdot x \cdot e^{2x} = \int \left(e^{2x} + x\right)dx$$

$$= \int e^{2x}dx + \int x dx$$

$$\therefore \quad y \cdot x \cdot e^{2x} = \frac{e^{2x}}{2} + \frac{x^2}{2} + C$$

$$\therefore \quad 2xye^{2x} = e^{2x} + x^2 + C_1 \qquad C_1 = 2C$$

Solved Example 40 :

$$\frac{dy}{dx} = \cos^3 x + y\cot x$$

Solution :

$$\frac{dy}{dx} - y\cot x = \cos^3 x$$

Here $P = -\cot x$,

$Q = \cos^3 x$

$$\text{I.F.} = e^{\int P dx} = e^{\int -\cot x} = e^{-\log\sin x}$$

$= \operatorname{cosec} x$

General solution is $y(\text{I. F}) = \int Q(\text{I.F}) + C$

$$y\operatorname{cosec} x = \int \cos^3 x(\operatorname{cosec} x)dx$$

$$= \int \frac{\cos^3 x}{\sin x}dx$$

Put $\sin x = t$, $\cos x\,dx = dt$

$$\int \frac{1-t^2}{t}dt = \int \frac{1}{t}dt - \int t dt = \log t - \frac{t^2}{2} + C$$

$$= \log|\sin x| - \frac{\sin^2 x}{2} + C$$

$$\therefore \quad y \cdot \operatorname{cosec} x = \log|\sin x| - \frac{\sin^2 x}{2} + C$$

$$\therefore \quad y = \sin x\left[\log|\sin x| - \frac{\sin^2 x}{2} + C\right]$$

1.8 First Order Differential Equations reducible to Linear form

An equation of the form

$$f'(y)\frac{dy}{dx}+P\,f(y)=Q,$$

where P and Q are constants or functions of x alone can be reduced to linear form as follows:

Putting f(y) = v so that $f'(y)\frac{dy}{dx}=\frac{dv}{dx}$,

Above equation becomes

$$\frac{dv}{dx}+Pv=Q$$

which is linear in v and x and its solution can be obtained by using working rule as for first order linear differential equation. Thus, we have I.F. = $e^{\int P\,dx}$

Solution is

$$\mathbf{v.e^{\int P\,dx} = \int Qe^{\int P\,dx}\,dx + c}$$

Finally, replacing v by f(y) will give solution in terms of x and y alone.

An equation of the form

$$f'(x)\frac{dx}{dy}+P_1\,f(x)=Q_1,$$

where P_1 and Q_1 are constants or functions of y alone can be reduced to linear form in the same way as describe above by putting f(x) = v.

Bernoulli's Equation

The equation of the form $\frac{dy}{dx}+Py=Qy^n$

where P and Q are the functions of x alone is called Bernoulli's equation.

When n = 0 or n = 1 it is already linear. For other values of n it can be reduced to linear equation by the substitution $z = y^{1-n}$. This is describe as follows:

Multiplying the above equation by y^{-n},

$$y^{-n}\frac{dy}{dx}+Py^{1-n}=Q$$

Let $y^{1-n} = z$

Diff. w.r.t. x, $(1-n)\, y^{-n} \cdot \dfrac{dy}{dx} = \dfrac{dz}{dx}$,

i.e., $y^{-n}\dfrac{dy}{dx} = \dfrac{1}{1-n}\dfrac{dz}{dx}$

The equation becomes

$$\frac{1}{1-n}\frac{dz}{dx} + Pz = Q \quad \text{or} \quad \frac{dz}{dx} + P(1-n)\,z = Q\,(1-n),$$

which is linear in z and x.

<u>Note :</u> $\dfrac{dx}{dy} + Px = Qx^n$ **is also in the Bernoulli's form. Here P and Q are functions of y alone. Method of solution is similar to that discussed above.**

Solved Example 41 :

Solve : $x\dfrac{dy}{dx} + y = y^2 \log x$

Solution :

Dividing equation by xy^2

$$y^{-2}\frac{dy}{dx} + \frac{1}{xy} = \frac{\log x}{x}$$

or $$y^{-2}\frac{dy}{dx} + y^{-1}\frac{1}{x} = \frac{\log x}{x}$$

Put $y^{-1} = t$

$$-1\, y^{-2}\frac{dy}{dx} = \frac{dt}{dx}$$

$$\therefore \quad -\frac{dt}{dx} + \frac{t}{x} = \frac{\log x}{x}$$

$$\therefore \quad \frac{dt}{dx} - \frac{t}{x} = -\frac{\log x}{x}$$

Here $P = -1/x$, $Q = -\dfrac{\log x}{x}$

$$\text{I.F.} = e^{\int P dx} = e^{\int -\frac{1}{x}dx} = e^{-\log x} = \frac{1}{x}$$

General solution is $t(\text{I.F}) = \int Q(\text{I.F})dx + C$

$$\therefore \quad t \cdot \frac{1}{x} = \int -\frac{\log x}{x}\cdot\frac{1}{x}dx$$

$$\therefore \quad \frac{t}{x} = \int -\frac{\log x}{x^2}dx$$

Integrating by parts on R.H.S.

$$\therefore \quad \frac{t}{x} = \log x \int -\frac{1}{x^2}dx - \int \frac{1}{x}\int -\frac{1}{x^2}dx$$

$$\frac{d}{dx}\log x = \frac{1}{x}$$

$$= \log x\left(\frac{1}{x}\right) - \int \frac{1}{x}\left(\frac{1}{x}\right)dx$$

$$\therefore \quad \frac{t}{x} = \frac{\log x}{x} + \frac{1}{x} + C$$

$$\therefore \quad y^{-1} = \log x + 1 + xC$$

$$\therefore \quad \frac{1}{y} = \log x + \log e + xC \qquad 1 = \log e$$

$$\therefore \quad \frac{1}{y} = \log(ex) + xC$$

Solved Example 42 :

Solve $x\frac{dy}{dx}+3y=x^4e^{1/x^2}\cdot y^3$

Solution :

Dividing by xy^3

$$y^{-3}\frac{dy}{dx}+\frac{3}{x}y^{-2}=x^3e^{1/x^2}$$

Put $y^{-2}=t$

$$\therefore\quad -2y^{-3}\frac{dy}{dx}=\frac{dt}{dx}$$

$$\therefore\quad -\frac{1}{2}\frac{dt}{dx}+\frac{3}{x}t=x^3e^{1/x^2}$$

$$\therefore\quad \frac{dt}{dx}-\frac{6}{x}t=-2x^3e^{1/x^2}$$

Here $P=-6/x$ $\qquad Q=-2x^3e^{1/x^2}$

$$\text{I.F.}=e^{\int Pdx}=e^{\int\frac{-6}{x}dx}=e^{-6\log x}=x^{-6}$$

General solution is t (I.F.) = $\int Q(\text{I.F.})dx+C$

$$tx^{-6}=\int -2x^3e^{\frac{1}{x^2}}x^{-6}dx$$

$$=\int -2x^3e^{1/x^2}x^{-6}dx$$

$$=-2\int e^{1/x^2}x^{-3}dx$$

Put $x^{-2}=u$

$$\therefore\quad -2x^{-3}dx=du$$

$$=\frac{-2}{-2}\int e^u du=e^{x^{-2}}$$

$$\therefore\quad tx^{-6}=e^{\frac{1}{x^2}}+C \quad\text{or}\quad y^{-2}x^{-6}=e^{\frac{1}{x^2}}+C$$

$$y^2x^6\left(e^{\frac{1}{x^2}}+C\right)=1$$

Solved Example 43 :

Solve $\frac{dy}{dx}+x\sin 2y=x^3\cos^2 y$

Solution :

Dividing by $\cos^2 y$

$$\sec^2 y\frac{dy}{dx}+2x\tan y=x^3$$

$$\sin 2A=2\sin A\cos A$$

Put t = tan y

$$\therefore\quad \frac{dt}{dx}=\sec^2 y\frac{dy}{dx}$$

$$\therefore\quad \frac{dt}{dx}+2xt=x^3$$

Here $P=2x$ $\qquad Q=x^3$

$$\text{I.F.}=e^{\int Pdx}=e^{\int 2xdx}=e^{x^2}$$

General solution is t(I.F) = $\int Q(\text{I.F})dx+C$

$$t\cdot e^{x^2}=\int x^3e^{x^2}dx$$

$$\int x^3e^{x^2}dx$$

Put $x^2=u$

$$2x\,dx=du\ \Rightarrow\ x\,dx=\frac{du}{2}=\frac{1}{2}\int ue^u du$$

Integrating by parts

$$=\frac{1}{2}\left[ue^u-\int\frac{du}{du}\left(\int e^u du\right)du\right]$$

$$=\frac{1}{2}\left[ue^u-e^u\right]$$

$$\text{Now}\quad t\cdot e^{x^2}=\frac{1}{2}\left[ue^u-e^u\right]+C$$

$$\tan y\cdot e^{x^2}=\frac{1}{2}\left[x^2\cdot e^{x^2}-e^{x^2}\right]+C$$

Solved Example 44 :

Solve : $\cos x \dfrac{dy}{dx} = y(\sin x - y)$

Solution :

Dividing the equation by $y^2 \cos x$

$$y^{-2}\frac{dy}{dx} - \frac{\tan x}{y} = -\sec x$$

$$\therefore \quad y^{-2}\frac{dy}{dx} - y^{-1}\tan x = -\sec x$$

Put $y^{-1} = t$

$$-y^{-2}\frac{dy}{dx} = \frac{dt}{dx}$$

$$-\frac{dt}{dx} - t\tan x = -\sec x$$

or $\dfrac{dt}{dx} + t\tan x = \sec x$

Here $P = \tan x$ $\quad Q = \sec x$

I.F. = $e^{\int P dx} = e^{\int \tan x dx} = e^{\log \sec x} = \sec x$

General solution is t (I.F.) = $\int Q(\text{I.F.}) + C$

$$t \cdot \sec x = \int (\sec x \sec x) dx$$

$$\therefore \quad t \cdot \sec x = \int \sec^2 x dx$$

$$\therefore \quad y^{-1}\sec x = \tan x$$

$$\therefore \quad \frac{1}{y}\sec x = \tan x + C$$

1.9 Differential Equations of first order and higher degree

A differential equation of first order and nth degree is of the form.

$$p^n + P_1 p^{n-1} + P_2 p^{n-2} + \ldots + P_{n-1} p + P_n = 0 \qquad \ldots(1)$$

where $p = \dfrac{dy}{dx}$ and $P_1, P_2, \ldots P_n$ are functions of x and y.

Equations solvable for p

The left hand side of (1) can be factorized into factors of the first degree then (1) becomes

$$(p - R_1)(p - R_2)\ldots(p - R_n) = 0$$

obtain a solution $f_i(x, y, c) = 0$ corresponding to the equation $p - R_i = 0$ for i = 1, 2, … n.

Thus the general solution of (1) is given by

$$f_1(x, y, c) f_2(x, y, c) \ldots f_n(x, y, c) = 0$$

Equations solvable for y

The equation can be put in the form y = f(x, p) …(2)

Differentiating w.r.t. x we get $p = \phi(x.\ p.\ \dfrac{dp}{dx})$ …(3)

which is a first order and first degree differential equation with variables p and x.

On solving equation (3) we get

$$\phi(x,p,c) = 0$$

Eliminating p from (1) and (3) we get the required solution.

Equations solvable for x

The equation can be put in the form x = f(y, p)

Differentiating w.r.t. y we get

$$\frac{1}{p} = \psi\left(y, p, \frac{dp}{dy}\right) \qquad \ldots(4)$$

This is first order and first degree differential equation with variables p and y.

On solving equation (4) we get

$$\phi(y,p,c) = 0 \qquad \ldots(5)$$

Then eliminating p from (1) and (5) we get the required solution.

Note : The factor which does not involve a derivative of p with respect to x or y will always lead to singular solution. Hence such a factor can be omitted.

Clairaut's Equation

The equation of the form

$$y = px + f(p) \qquad \ldots(1)$$

is called Clairaut's equation.

Differentiating w.r.t. x we get $p = p + \left(x + f'(p)\right)\left(\frac{dp}{dx}\right)$

$\therefore \quad x + f'(p) = 0 \quad$ or $\quad \frac{dp}{dx} = 0$

Now $\quad \frac{dp}{dx} = 0$

$\therefore \quad p = c \qquad$ (a constant)

Hence the general solution of (1) is

$$\text{G.S.} = y = cx + f(c) \qquad \ldots(2)$$

if x + f′(p) = 0 we use equation (2) and (1) to obtain a solution. This solution is not included in the general solution (2). Such a solution is called a singular solution.

Note : Clairaut's equation always has a singular solution

Solved Example 45 :

Solve $p^2 - 5p + 6 = 0$

Solution :

Solving for p

$\therefore \quad p^2 - 3p - 2p + 6 = 0$

$\therefore \quad p(p-3) - 2(p-3) = 0$

$\therefore \quad (p-3)(p-2) = 0$

$\therefore \quad p = 3$ and $p = 2$

when $p = 3$, $\dfrac{dy}{dx} = 3$

$\therefore \quad y = 3x + C$ [Integrating]

when $p = 2$, $\dfrac{dy}{dx} = 2$

$\therefore \; y = 2x + C$ [Integrating]

$\therefore$ The solution is $(y - 3x - C)\,(y - 2x - C)$

Solved Example 46 :

Solve $xp^2 - 2py + x = 0$

Solution :

Solving for p

$$p = \frac{2y \pm \sqrt{4y^2 - 4x^2}}{2x}$$

$$\therefore \quad \frac{dy}{dx} = \frac{y \pm \sqrt{y^2 - x^2}}{x}$$

This is a homogenous equation in x and y

Put y = vx

$$\therefore \quad \frac{dy}{dx} = v + x\frac{dv}{dx}$$

$$\therefore \quad v + x\frac{dv}{dx} = v \pm \sqrt{v^2 - 1}$$

$$\therefore \quad x\frac{dv}{dx} = \sqrt{v^2 - 1}$$

Separating the variables we get

$$\frac{dv}{\sqrt{v^2 - 1}} = \frac{dx}{x}$$

$$\therefore \quad \log\left(v + \sqrt{v^2 - 1}\right) = \log x + C$$

$$\therefore \quad \log\left(\frac{y}{x} + \sqrt{\frac{y}{x} - 1}\right) - \log x = \log C_1$$

$C = \log C_1$

$$\therefore \quad \log\left(\frac{\frac{y}{x} + \sqrt{\frac{y-x}{x}}}{x}\right) = \log C_1$$

$$\therefore \quad \frac{xy + \sqrt{y - x}}{x^3} = C_1$$

Solved Example 47 :

Solve $3x - y + \log p = 0$

Solution :

$y = 3x + \log p$ solving for y

Differentiating w.r.t. x

$$p = 3 + \frac{1}{p}\frac{dp}{dx}$$

$$\therefore \quad p^2 - 3p = \frac{dp}{dx}$$

$$\therefore \quad p(p-3) = \frac{dp}{dx}$$

$$\therefore \quad dx = \frac{dp}{p(p-3)} = \frac{1}{3}\left(\frac{1}{p-3} - \frac{1}{p}\right)dp$$

[partial fractions]

$$\therefore \quad 3dx = \left(\frac{1}{p-3} - \frac{1}{p}\right)dp$$

Integrating we get

$$3x = \log(p-3) - \log p - \log C$$

$$\therefore \quad 3x = \log\frac{(p-3)}{Cp}$$

$$\frac{p-3}{p} = Ce^{3x} \qquad \text{then} \qquad 1-\frac{3}{p} = Ce^{3x}$$

$$\therefore \quad \frac{3}{p} = 1 - Ce^{3x}$$

$$\therefore \quad p = \frac{3}{1-Ce^{3x}}$$

Eliminating p from the given equation

$$y = 3x + \log\left(\frac{3}{1-Ce^{3x}}\right)$$

Solved Example 48 :

Solve $y = 2px + y^2 p^3$

Solution :

Solving for x

$$x = \left(\frac{y}{2p} - \frac{y^2p^2}{2}\right)$$

Differentiating w.r.t. y we get

$$\frac{1}{p} = \frac{1}{2p} - \frac{y}{2p^2}\left(\frac{dp}{dy}\right) - yp^2 - py^2\left(\frac{dp}{dy}\right)$$

$$\therefore \quad \frac{1}{2p} + p^2y + \left(\frac{y}{2p^2} + py^2\right)\frac{dp}{dy} = 0$$

$$\therefore \quad \left(\frac{1+2p^3y}{2p}\right) + y\left(\frac{1+2p^3y}{2p^2}\right)\frac{dp}{dy} = 0$$

$$\therefore \quad \left(\frac{1+2p^3y}{2p}\right)\left(1 + y\frac{dp}{pdy}\right) = 0$$

Taking $1 + \frac{y}{p}\frac{dp}{dy} = 0$

$$\therefore \quad \frac{dy}{y} + \frac{dp}{p} = 0$$

$$\therefore \quad \log y + \log p = \log c$$

$$\therefore \quad py = c \qquad \therefore \quad p = c/y$$

Substituting in given equation

$$\therefore \quad y = 2\frac{c}{y}x + y^2\left(\frac{c}{y}\right)^3$$

Solved Example 49 :

Solve $y = px + \frac{a}{p}$ and obtain the singular solution

Solution :

Differentiating the equation w.r.t. x we get

$$p = p + x\frac{dp}{dx} - \frac{a}{p^2}\left(\frac{dp}{dx}\right)$$

$$\therefore \quad \frac{dp}{dx}\left(x - \frac{a}{p^2}\right) = 0$$

$$\therefore \quad \frac{dp}{dx} = 0 \qquad \therefore \quad p = c$$

$$\text{or} \quad x = \frac{a}{p^2} \qquad \therefore \; p = \sqrt{\frac{a}{x}}$$

Substituting p = C in the given equation

The general solution is given by

$$y = cx + \frac{a}{c}$$

Substituting $p = \sqrt{\frac{a}{x}}$ in the given equation

The singular solution is given by

$$y = x\sqrt{\frac{a}{x}} + a\sqrt{\frac{x}{a}} = 2\sqrt{ax} \qquad \text{or} \quad y^2 = 4ax$$

Solved Example 50 :

Solve cos y sin px = sin y cos px + p

Solution :

The given equation is $\sin(px - y) = p$

$\therefore \quad px - y = \sin^{-1} p$

$\therefore \quad y = px - \sin^{-1} p$ [Clairaut's form]

General solution is $y = Cx - \sin^{-1} C$

Solved Example 51 :

Solve $(px - y)(1 - p) = p$.

Solution :

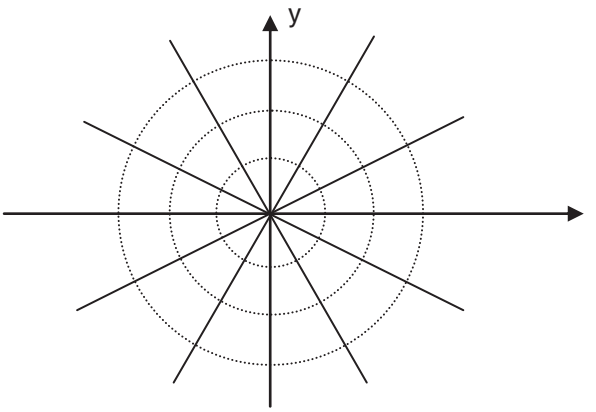
The equation is $(y - px)(p - 1) = p$

$\therefore \quad y(p-1) = p + px(p-1)$

$\therefore \quad y = px + \dfrac{p}{p-1}$ [Clairaut's form]

General solution is $y = Cx + \dfrac{C}{C-1}$

1.10 Orthogonal trajectories of family of curves



The general solution of a first order differential equation represents a parameter family of curves. Two curves intersect orthogonally if the angle of intersection between the curves is a right angle.

<u>For Example</u> :

Concentric circles $x^2 + y^2 = r^2$ with centre at origin are orthogonal to the radial lines y = mx

Differentiating $x^2 + y^2 = r^2$

slope = $m_1 = \dfrac{dy}{dx} = -\dfrac{x}{y}$

slope of straight line $m_2 = \dfrac{y}{x}$

$m_1 m_2 = -1$

Problem Formulation

Let f(x, y, c) = 0 be a parameter family of curves. Differentiating and eliminating c. we get, differential equation for which f(x, y, c) = 0 are the integral curves.

Let Differential equation be F(x, y, y′) = 0 and slope of integral curves f(x, y, c) = 0 is y′

∴ Slope of curve orthogonal to it is –1/y′.

Hence the differential equation governing the family of orthogonal curves is

$$F\left(x, y, -\frac{1}{y'}\right) = 0$$

Solved Example 52 :

Obtain the orthogonal trajectories of the hyperbolas $x^2 - y^2 = c$

Solution :

Differentiating the above equation

$$2x - 2yy' = 0$$

This is the differential equation governing the family of hyperbolas.

The differential equation governing the orthogonal trajectories are

$$2x - 2y\left(-\frac{1}{y'}\right) = 0$$

or $xy' + y = 0$

or $xdy + ydx = 0$

or $d(xy) = 0$

or $xy = C$

hence, the rectangular hyperbolas xy = C are the orthogonal trajectories of the hyperbolas $x^2 - y^2 = C$

Solved Example 53 :

Show that the one parameter family of the curves $y^2 = 4C(C + x)$ are self orthogonal.

Solution :

Differentiating $y^2 = 4C(C + x)$ we get

$$2yy' = 4C$$

$$\therefore \quad C = \frac{yy'}{2}$$

Substituting in given equation

$$y^2 = 4\left(\frac{yy'}{2}\right)\left[\frac{yy'}{2} + x\right] = 2yy'\left[\frac{yy' + 2x}{2}\right]$$

$$\therefore \quad y^2 = yy'[yy' + 2x] \qquad \ldots(1)$$

This is the differential equation governing the given family

Replace y′ by –1/y′ we get the differential equation governing the orthogonal family as

$$y^2 = -\frac{y}{y'}\left[-\frac{y}{y'} + 2x\right] = \frac{y^2 - 2xyy'}{(y')^2}$$

$$y^2 = y^2(y')^2 + 2xyy'$$

i.e. $y^2 = yy'(yy' + 2x) \qquad \ldots(2)$

Equation (1) and (2) are same.

Hence, the orthogonal trajectories are the given curves themselves.

1.11 Higher Order Linear Differential Equations with constant coefficients

A linear differential equation of nth order, with constant coefficients is of the form

$$\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \ldots + a_n y = X \qquad \ldots(1)$$

where $a_1, a_2, a_3 \ldots a_n$ are constants and X is a function of x.

Thus, $\left(D^n + a_1 D^{n-1} + a_2 D^{n-2} + \ldots + a_n\right) y = X$

where $D = \frac{d}{dx}$,

$$D^2 = \frac{d^2}{dx^2}$$

$$D^n = \frac{d^n}{dx^n}$$

Consider $\left(D^n + a_1 D^{n-1} + a_2 D^{n-2} + \ldots + a_n\right) y = 0 \qquad \ldots(2)$

The general solution of (2) is given by

$$Y = c_1 y_1 + c_2 y_2 + \ldots + c_n y_n$$

where $y_1, y_2 \ldots y_n$ are n independent solutions and $c_1, c_2, \ldots, c_n$ are arbitrary constants.

Y is called the complementary function (C.F.) of equation (1)

Let u is a particular solution (particular integral) of equation (1). Then general solution of equation (1) is of the form.

$$\mathbf{y = y_c + y_p}$$

where y_c is complementary function (C.F.)

y_p is particular integral (P.I)

Methods of finding Complementary Function

Let the differential equation be

$$f(D)\, y = X$$

We write f(D) = 0. This algebraic equation in D is called as Auxiliary Equation. We solve for D and find the roots.

Let $m_1, m_2, \ldots, m_n$ be the roots.

Depending upon the nature of the roots, we have four cases.

Case	Roots	Complementary Function
1.	Real and distinct	$y = C_1e^{m_1x} + C_2e^{m_2x} + \ldots + C_ne^{m_nx}$
2.	Real and repeated	m_1 be repeated two times m_2 be repeated three times $y = (C_1x + C_2)e^{m_1x} + (C_3x^2 + C_4x + C_5)e^{m_2x} + \ldots$
3.	Complex roots	$\left.\begin{array}{l} m_1 = \alpha + i\beta \\ m_2 = \alpha - i\beta \end{array}\right\}$ Conjugate Pair $y = e^{\alpha x}(C_1 \cos\beta x + C_2 \sin\beta x)$
4.	Repeated complex roots	$m_1 = \alpha + i\beta$ $m_2 = \alpha - i\beta$ $m_3 = \alpha + i\beta$ $m_4 = \alpha - i\beta$ $y = e^{\alpha x}\left[(C_1x + C_2)\cos\beta x + (C_3x + C_4)\sin\beta x\right] + \ldots$

Solved Example 54 :

Find complementary function for the following :

$y''' - 9y' = 0$

Solution :

Let $D = \dfrac{d}{dx}$; then the equation becomes

$(D^3 - 9D)y = 0$

Auxiliary Equation is $D^3 - 9D = 0$

$\therefore \quad D(D^2 - 9) = 0$

$\therefore \quad D(D^2 - 3^2) = 0$

$\therefore \quad D(D-3)(D+3) = 0$

$\therefore \quad D = 0, 3, -3$

Roots are real and distinct.

C.F. is $y = C_1 + C_2 e^{3x} + C_3 e^{-3x}$

Solved Example 55 :

Find complementary function for

$3y''' - 2y'' - 3y' + 2y = 0$

Solution :

Let $D = \dfrac{d}{dx}$ then equation becomes

$(3D^3 - 2D^2 - 3D + 2)y = 0$

A.E. is $3D^3 - 2D^2 - 3D + 2 = 0$

1	3	–2	–3	2
	0	3	1	–2
	3	1	–2	0

$(D-1)(3D^2 + D - 2) = 0$

$\therefore \quad (D-1)\left[3D(D+1) - 2(D+1)\right] = 0$

$\therefore \quad (D-1)(3D-2)(D+1)=0$

$\therefore \quad D = 1, 2/3, -1$

Roots are real and distinct.

C.F. is $y = C_1e^x + C_2e^{2x/3} + C_3e^{-x}$

Solved Example 56 :

Find complementary function for $y''' + y' = 0$

Solution :

Let $D = \dfrac{d}{dx}$ then equation becomes

$(D^3 + D)y = 0$

A.E. is $D^3 + D = 0$

$\therefore \quad D(D^2+1) = 0$

$\therefore \quad D = 0, \pm i \qquad \sqrt{-1} = i$

Roots are

$0 \rightarrow$ real

$\pm i \rightarrow$ complex roots

C.F. is $y = C_1 + C_2\cos x + C_3\sin x$

Solved Example 57 :

Find complementary function for

$y''' + 4y'' + 5y' + 2y = 0$

Solution :

Let $D = \dfrac{d}{dx}$ then equation becomes

$(D^3 + 4D^2 + 5D + 2)y = 0$

A.E. is $D^3 + 4D^2 + 5D + 2 = 0$

-1	1	4	5	2
	0	-1	-3	-2
	1	3	2	0

$(D+1)(D^2+3D+2) = 0$

$\therefore \quad (D+1)(D^2+2D+D+2) = 0$

$\therefore \quad (D+1)(D+1)(D+2) = 0$

$\therefore \quad D = -1, -1, -2$

C.F. is $y = (C_1x + C_2)e^{-x} + C_3e^{-2x}$

Solved Example 58 :

Find complementary function for

$y''' - 2y'' + y' = \sin 2x$

Solution :

Let $D = \dfrac{d}{dx}$ then equation becomes

$(D^3 - 2D^2 + D)y = \sin 2x$

A.E. is $D^3 - 2D^2 + D = 0$

$\therefore \quad D(D^2 - 2D + 1) = 0$

$\therefore \quad D(D-1)(D-1) = 0$

$\therefore \quad D = 0, 1, 1$

C.F. is $y = C_1 + (C_2x + C_3)e^x$

Solved Example 59 :

Find complementary function for

$\dfrac{d^4y}{dx^4} - 81y = 0$

Solution :

Let $D = \dfrac{d}{dx}$ then equation becomes

$(D^4 - 81)y = 0$

A.E. is $(D^2 - 3^2)(D^2 + 3^2)$

$\therefore \quad (D-3)(D+3)(D+3i)(D-3i) = 0$

$\therefore \quad D = \pm 3, \pm 3i$

C.F. is

$$y = C_1e^{3x} + C_2e^{-3x} + C_3\cos 3x + C_4\sin 3x$$

Solved Example 60 :

Find complementary function for

$$y''' - 6y'' + 11y' - 6y = 0$$

Solution :

Let $D = \frac{d}{dx}$ then equation becomes

$$(D^3 - 6D^2 + 11D - 6)y = 0$$

A.E. is $D^3 - 6D^2 + 11D - 6 = 0$

$\therefore \quad (D-1)(D-2)(D-3) = 0$

$\therefore \quad D = 1, 2, 3$

C.F. is $y = C_1e^x + C_2e^{2x} + C_3e^{3x}$

Solved Example 61 :

Find complementary function for

$y''' + y'' - 3y' + y = 0$

Solution :

Let $D = \frac{d}{dx}$ then equation becomes

$$\left(D^3 + D^2 - 3D + 1\right)y = 0$$

A.E. is $\left(D^3 + D^2 - 3D + 1\right) = 0$

$\therefore \quad (D-1)\left(D^2 + 2D - 1\right) = 0$

$\therefore \quad D = 1, -1 \pm \sqrt{2}$

C.F. is $y = C_1e^x + C_2e^{\left(-1+\sqrt{2}\right)x} + C_3e^{\left(-1-\sqrt{2}\right)x}$

Method of finding Particular Integral

Consider $\left(D^n + a_1D^{n-1} + \ldots + a_n\right)y = X$ …(1)

This equation can be written as f(D)y = X where

$$f(D) = D^n + a_1D^{n-1} + \ldots + a_n$$

we now introduce the operator $\frac{1}{f(D)}$.

Definition

$\left[\frac{1}{f(D)}\right]X$ is a function of x free from arbitrary constant s which when operated by f(D) gives X.

If $\left[\frac{1}{f(D)}\right]X = u$ then $\left[f(D)\right](u) = X$. Hence u is a particular solution of (1)

From the definition we get $f(D)\left[\frac{1}{f(D)}\right]X = X$ so that $\frac{1}{f(D)}$ is the inverse operator of f(D).

In particular $\frac{1}{D}$ stands for integration.

General Method :

$$\frac{1}{D-m}X = e^{mx}\int e^{-mx}Xdx$$

$$\frac{1}{D+m}X = e^{-mx}\int e^{mx}Xdx$$

Note : **While finding P.I. arbitrary constants are not considered.**

Particular integral for some commonly used functions

Type 1 :

When f(a) $\neq$ 0

$$\frac{1}{f(D)}e^{ax} = \frac{1}{f(a)}e^{ax}$$

When f(a) = 0

$$\frac{1}{f(D)}e^{ax} = \frac{1}{\phi(a)}\frac{x^r e^{ax}}{r!} \quad \phi(a) \neq 0$$

Type 2a :

When f($-a^2$) $\neq$ 0

$$\frac{1}{f(D^2)}\sin(ax+b) = \frac{1}{f(-a^2)}\sin(ax+b)$$

When f($-a^2$) = 0

$$\frac{1}{f(D^2)}\sin(ax+b) = \frac{1}{\phi(-a^2)}\frac{(-1)^r x^r}{r!(2a)^r}\sin\left(ax+b+\frac{r\pi}{2}\right)$$

Type 2b :

When f (a^2) $\neq$ 0

$$\frac{1}{f(D^2)}\sinh(ax+b) = \frac{1}{f(a^2)}\sinh(ax+b)$$

Type 3 :

$$\frac{1}{f(D)}x^m$$

Expand $\frac{1}{f(D)}$ in ascending powers of D as far as D^m and operate on x^m

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots$$

Type 4 :

$$\frac{1}{f(D)}e^{ax}V = e^{ax}\frac{1}{f(D+a)}V$$

where V is any function of x.

Type 5 :

$$\frac{1}{f(D)}xV = \left[x - \frac{1}{f(D)}f'(D)\right]\frac{1}{f(D)}V$$

where v is any function of x and $f'(D)$ is the expression obtained by differentiating f(D) with respect to D.

Following formulae will be useful while finding particular integral in some cases

$$(1 + D)^{-1} = 1 - D + D^2 - D^3 + \ldots..$$
$$(1 - D)^{-1} = 1 + D + D^2 + D^3 + \ldots..$$

Solved Example 62 :

Find P.I. of the following

$(D^2 + 3D + 2)y = \sin e^x$

Solution :

P.I. (particular integral) is

$$y = \frac{1}{(D+1)(D+2)}\sin e^x$$

$$= \left[\frac{1}{D+1} - \frac{1}{D+2}\right]\sin e^x \quad \text{By Partial fraction}$$

$$= \frac{1}{D+1}\sin e^x - \frac{1}{D+2}\sin e^x$$

$$= e^{-x}\int e^x \sin e^x dx - e^{-2x}\int e^{2x}\sin e^x dx$$

Put $e^x = t$

$\therefore \quad e^x dx = dt$

$$\therefore \quad y = e^{-x}\int \sin t\, dt - e^{-2x}\int t \sin t\, dt$$

$$= e^{-x}(-\cos t) - e^{-2x}\left[t(-\cos t) - 1(-\sin t)\right]$$

$$= -e^{-x}\cos e^x - e^{-2x}\left[-e^x \cos e^x + \sin e^x\right]$$

$$= -e^{-2x}\sin e^x$$

PROBLEMS BASED ON TYPE 1

Solved Example 63 :

Solve $(D-2)^2 y = e^{2x}$

Solution :

Auxillary equation is

$$(D-2)^2 = 0$$

$\therefore \quad D = 2, 2$

complementary function (C.F) is

$$y = (C_1x + C_2)e^{2x}$$

(P.I) Particular integral is

$$y = \frac{1}{(D-2)^2}e^{2x} = \frac{x^2}{2!}e^{2x} = \frac{x^2e^{2x}}{2}$$

G.S. is y = C.F. + P. I.

$$= (C_1x + C_2)e^{2x} + \frac{x^2e^{2x}}{2}$$

Solved Example 64 :

Solve $y'' - 2y' - 3y = 3e^{2x}$

Solution :

Let $D = \dfrac{d}{dx}$

A.E. is $D^2 - 2D - 3 = 0$

$\therefore \quad D^2 - 3D + D - 3 = 0$

$\therefore \quad (D-3)(D+1) = 0$

$\therefore \quad D = 3, -1$

C.F. is $y = C_1e^{3x} + C_2e^{-x}$

P.I. is $y = \dfrac{1}{(D-3)(D+1)}3e^{2x}$

$$= 3\frac{1}{(2-3)(2+1)}e^{2x} = -e^{2x}$$

G.S. is $y = C_1e^{3x} + C_2e^{-x} - e^{2x}$

Solved Example 65 :

Find General solution of $y''' - 2y'' - 5y' + 6y = 4e^{-x} - e^{2x}$

Solution :

Let $D = \dfrac{d}{dx}$

$\therefore$ A.E. is $D^3 - 2D^2 - 5D + 6 = 0$

3	1	–2	–5	6
	0	3	3	–6
	1	1	–2	0

$$(D-3)(D^2 + D - 2) = 0$$

$\therefore \quad (D-3)(D^2 + 2D - D - 2) = 0$

$\therefore \quad (D-3)(D-1)(D+2) = 0$

$\therefore \quad D = 3, 1, -2$

C.F. is $y = C_1e^{3x} + C_2e^{x} + C_3e^{-2x}$

P.I. is $y = \dfrac{1}{(D-3)(D-1)(D+2)}4e^{-x} - e^{2x}$

$$= 4\frac{1}{(D-3)(D-1)(D+2)}e^{-x}$$

$$-\frac{1}{(D-3)(D-1)(D+2)}e^{2x}$$

$$= 4\frac{1}{(-1-3)(-1-1)(-1+2)}e^{-x}$$

$$-\frac{1}{(2-3)(2-1)(2+2)}e^{2x}$$

$$= \frac{1}{2}e^{-x} + \frac{1}{4}e^{2x}$$

$$= \frac{e^{-x}}{2} + \frac{1}{4}e^{2x}$$

G.S. is

$$y = C_1e^{3x} + C_2e^{x} + C_3e^{-2x} + \frac{e^{-x}}{2} + \frac{1}{4}e^{2x}$$

Solved Example 66 :

Evaluate P.I. of $y'' + y' - 6y = 5e^{-3x}$

Solution :

P.I. is

$$y = \frac{1}{\left(D^2 + D - 6\right)} 5e^{-3x}$$

$$= 5 \cdot \frac{1}{(D+3)(D-2)} e^{-3x}$$

$$= 5 \cdot \frac{1}{(D+3)} \left[\frac{1}{-3-2}\right] e^{-3x}$$

$$= 5 \cdot x \frac{1}{(-5)} e^{-3x} \qquad \left[\frac{1}{f(D)} e^{ax} = \frac{1}{\phi(a)} \frac{x^r e^{ax}}{r!}\right]$$

$$= -xe^{-3x}$$

Solved Example 67 :

Evaluate the G.S. of $4\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + y = e^{x/2}$

Solution :

Let $D = \frac{d}{dx}$

A.E. is $\left(4D^2 - 4D + 1\right) = 0$

$\therefore \quad (2D-1)^2 = 0$

$\therefore \quad D = \frac{1}{2}, \frac{1}{2}$

C.F. is $\quad y = \left(C_1x + C_2\right)e^{x/2}$

P.I. is $\quad y = \frac{1}{(2D-1)^2} e^{x/2} = \frac{1}{4\left(D - \frac{1}{2}\right)^2} e^{x/2}$

$$= \frac{1}{4} \frac{x^2}{2!} e^{x/2}$$

$$\left[\frac{1}{f(D)} e^{ax} = \frac{1}{\phi(a)} \frac{x^r e^{ax}}{r!}\right]$$

$$= \frac{x^2 e^{x/2}}{8}$$

G.S. is $\quad y = \left(C_1x + C_2\right)e^{x/2} + \frac{x^2e^{x/2}}{8}$

Solved Example 68 :

Evaluate P.I. of $\left(D^2 + 4D + 4\right)y = 2^x$

Solution :

P.I. is $\quad y = \frac{1}{(D+2)^2} 2^x$

Now $\quad 2^x = e^{\log 2^x} = e^{x\log 2} = e^{Cx} \qquad C = \log 2$

$$\therefore \quad y = \frac{1}{(D+2)^2} e^{cx}$$

$$= \frac{1}{(C+2)^2} e^{cx}$$

$$= \frac{1}{(\log 2 + 2)^2} 2^x$$

Solved Example 69 :

Solve $y'' + 4y' + 13y = 2e^{-x}$ given $y(0) = 0$ and $y'(0) = -1$

Solution :

A.E. is

$$D^2 + 4D + 13 = 0$$

$$\therefore \quad D = \frac{-4 \pm \sqrt{16-52}}{2}$$

$$= \frac{-4 \pm \sqrt{-36}}{2}$$

$$= -2 \pm 3i$$

C.F. is $\quad y = e^{-2x}\left(C_1 \cos 3x + C_2 \sin 3x\right)$

P.I. is $\quad y = \left(\frac{1}{D^2 + 4D + 13}\right) 2e^{-x}$

$$= 2\left(\frac{1}{1-4+13}\right) e^{-x}$$

$$= \frac{1}{5} e^{-x}$$

G.S. is

y = C.F. + P.I.

$$= e^{-2x}\left(C_1 \cos 3x + C_2 \sin 3x\right) + \frac{1}{5} e^{-x}$$

Now y(0) = 0

$\therefore \quad C_1 = -1/5$

$$y' = e^{-2x}\left[-3C_1 \sin 3x + 3C_2 \cos 3x\right]$$

$$-2e^{-2x}\left(C_1 \cos 3x + C_2 \sin 3x\right) - \frac{1}{5} e^{-x}$$

Now y′(0) = –1

$$\therefore \quad -1 = 3C_2 - 2C_1 - \frac{1}{5}$$

Hence $\quad C_2 = -\frac{2}{5}$

$$\therefore \quad y = -\frac{1}{5} e^{-2x}\left(\cos 3x + 2\sin 3x\right) + \frac{1}{5} e^{-x}$$

PROBLEMS BASED ON TYPE 2 :

Solved Example 70 :

P.I. of $\frac{1}{D^2+9} 4\sin 3x$

Solution :

P.I. is

$$y = \frac{1}{D^2+9} 4\sin 3x$$

$$= 4 \cdot \frac{1}{D^2+9} \sin 3x$$

$$= 4 \cdot \frac{(-1)^1 x^1}{1!(2)(3)} \cos 3x$$

$$\left[\left(\sin 3x + 90\right) = \cos 3x\right]$$

$$= \frac{-2x \cos 3x}{3}$$

Solved Example 71 :

Evaluate P.I. of $\left(D^3 - 1\right) y = \sin 4x$

Solution :

P.I. is

$$y = \frac{1}{\left(D^3 - 1\right)} \sin 4x = \frac{1}{D \cdot D^2 - 1} \sin 4x$$

$$= \frac{1}{D\left(-4^2\right) - 1} \sin 4x = \frac{1}{D(-16) - 1} \sin 4x$$

$$= \frac{-1}{16D + 1} \sin 4x = \frac{-(16D - 1)}{(16D + 1)(16D - 1)} \sin 4x$$

$$= \frac{-(16D - 1)}{(256)(-16) - 1} \sin 4x = \frac{16D - 1}{4097} \sin 4x$$

$$= \frac{1}{4097} 16 \cdot D \sin 4x - \frac{1}{4097} \sin 4x$$

$$= \frac{64}{4097} \cos 4x - \frac{\sin 4x}{4097}$$

Solved Example 72 :

Evaluate P.I. of $y'' - 5y' + 4y = 65 \sin 2x$

Solution :

P.I. is

$$y = \frac{1}{D^2 - 5D + 4} 65 \sin 2x$$

$$= 65 \rangle \frac{1}{-4 - 5D + 4} \sin 2x$$

$$= 65 \cdot \frac{1}{-5D} \sin 2x = \frac{13}{2} \cos 2x$$

Solved Example 73 :

Evaluate P.I. of $y''' - y'' + 4y' - 4y = \sin 3x$

Solution :

P.I is $\quad y = \dfrac{1}{D^3 - D^2 + 4D - 4} \sin 3x$

By synthetic division

1	1	–1	4	–4
	0	1	0	4
	1	0	4	$\boxed{0}$

$\therefore \quad D^3 - D^2 + 4D - 4 = (D - 1)(D^2 + 4)$

$$\therefore \quad y = \frac{1}{(D-1)(D^2+4)} \sin 3x$$

$$= \frac{1}{-3^2 + 4} \frac{1}{D-1} \sin 3x$$

$$= \frac{1}{-5} \frac{1}{(D-1)} \sin 3x = -\frac{1}{5} \frac{(D+1)}{D^2 - 1} \sin 3x$$

$$= -\frac{1}{5} \frac{1}{-3^2 - 1} (D+1) \sin 3x$$

$$= \frac{1}{50} [3 \cos 3x + \sin 3x]$$

Solved Example 74 :

P.I. of

$$(D^5 - D^4 + 2D^3 - 2D^2 + D - 1)y = \cos x$$

Solution :

P.I. is

$$y = \frac{1}{D^5 - D^4 + 2D^3 - 2D^2 + D - 1} \cos x$$

By synthetic division of

$D^5 - D^4 + 2D^3 - 2D^2 + D - 1$ we get

Sum of coefficients is zero

$\therefore \quad (D - 1)$ is a factor

1	1	–1	2	–2	1	–1
	0	1	0	2	0	1
	1	0	2	0	1	$\boxed{0}$

$$(D-1)(D^4 + 2D^2 + 1)$$

or $\quad (D-1)(D^2+1)^2$

$$\therefore \quad y = \frac{1}{(D-1)(D^2+1)^2} \cos x$$

$$\therefore \quad y = \frac{(D+1)}{(D^2-1)(D^2+1)^2} \cos x$$

$$= \frac{1}{(D^2+1)^2} (D+1) \left[\frac{1}{(D^2-1)} \cos x \right]$$

$$= \frac{1}{(D^2+1)^2} (D+1) \left[-\frac{1}{2} \cos x \right]$$

$$= -\frac{1}{2} \frac{1}{(D^2+1)^2} [(D+1) \cos x]$$

$$= -\frac{1}{2}\frac{1}{\left(D^2+1\right)^2}\left[\cos x - \sin x\right]$$

$$= -\frac{1}{2}\left[\frac{1}{\left(D^2+1\right)^2}\cos x\right]$$

$$+\frac{1}{2}\frac{1}{\left(D^2+1\right)^2}\sin x$$

$$= -\frac{1}{2}\left[\frac{(-1)^2 x^2}{2!2^2}\cos\left(x+\frac{2\pi}{2}\right)\right]$$

$$+\frac{1}{2}\left[\frac{(-1)^2 x^2}{2!2^2}\sin\left(x+\frac{2\pi}{2}\right)\right]$$

$$= \frac{x^2}{16}\cos x - \frac{x^2}{16}\sin x$$

$$= \frac{x^2}{16}\left(\cos x - \sin x\right)$$

Solved Example 75 :

Evaluate P.I. of $\frac{d^6y}{dx^6} + y = \cos x$

Solution :

P.I. of $y = \frac{1}{D^6+1}\cos x$

$$= \frac{1}{\left(D^2+1\right)\left(D^4-D^2+1\right)}\cos x$$

$$= \left(\frac{1}{D^2+1}\right)\left(\frac{1}{(-1)^2-(-1)+1}\right)\cos x$$

$$= \frac{1}{3}\frac{1}{D^2+1}\cos x = \frac{1}{3}\frac{x\sin x}{2}$$

Solved Example 76 :

Solve $\left(D^2+D+1\right)y = \sin 2x$

Solution :

A.E. is $D^2+D+1=0$

$$\therefore \quad D = \frac{-1\pm\sqrt{1-4\times 1}}{2} = \frac{-1\pm i\sqrt{3}}{2}$$

C.F. is $y = e^{-x/2}\left(C_1\cos\frac{\sqrt{3}}{2}x + C_2\sin\frac{\sqrt{3}}{2}x\right)$

P.I. is $y = \frac{1}{D^2+D+1}\sin 2x$

$$= \frac{1}{-2^2+D+1} = \frac{1}{D-3}\sin 2x$$

$$= \frac{D+3}{D^2-9}\sin 2x$$

$$= \frac{D+3}{-2^2-9}\sin 2x = \frac{-1}{13}(D+3)\sin 2x$$

$$= \frac{-1}{13}\left[2\cos 2x + 3\sin 2x\right]$$

G.S. is

y = C.F. + P.I.

$$\therefore \quad y = e^{-x/2}\left(C_1\cos\frac{\sqrt{3}}{2}x + C_2\sin\frac{\sqrt{3}}{2}x\right)$$

$$-\frac{1}{13}\left[2\cos 2x + 3\sin 2x\right]$$

PROBLEMS BASED ON TYPE 3

Solved Example 77 :

Evaluate P.I. of $\left(D^2+3D+2\right)y = x^2$

Solution :

P.I. is

$$y = \frac{1}{D^2+3D+2}x^2$$

$$= \frac{1}{2}\left(1+\frac{3D+D^2}{2}\right)^{-1} x^2$$

$$= \frac{1}{2}\left[1-\left(\frac{3D+D^2}{2}\right)+\left(\frac{3D+D^2}{2}\right)^2\right]x^2$$

$$\left[(1+x)^{-1} = 1-x+x^2-x^3+x^4+....\right]$$

$$= \frac{1}{2}\left(1-\frac{3D}{2}+\frac{7D^2}{4}\right)x^2 = \frac{1}{2}\left(x^2-3x+\frac{7}{2}\right)$$

Solved Example 78 :

Find G.S. of $(D^2+5D+4)y = x^2+7x+9$

Solution :

A.E. is $(D^2+5D+4)=0$

$\therefore \quad D^2 + 4D + D + 4 = 0$

$\therefore \quad D(D+4)+1(D+4)=0$

$\therefore \quad (D+1)(D+4)=0$

$\therefore \quad D = -1, -4$

C.F. is $y = C_1e^{-x}+C_2e^{-4x}$

P.I. is

$$y = \frac{1}{D^2+5D+4}x^2+7x+9$$

$$= \frac{1}{4\left(1+\frac{D^2}{4}+\frac{5D}{4}\right)}x^2+7x+9$$

$$= \frac{1}{4}\left(1+\frac{D^2}{4}+\frac{5D}{4}\right)^{-1} x^2+7x+9$$

$$= \frac{1}{4}\left[1-\frac{D^2}{4}-\frac{5D}{4}+\left(\frac{D^2}{4}+\frac{5D}{4}\right)^2\right](x^2+7x+9)$$

$$\ldots\left[(1+x)^{-1} = 1-x+x^2-x^3+x^4+\ldots\right]$$

Neglecting coefficients of D^4, D^3

$$= \frac{1}{4}\left[1-\frac{5D}{4}+\frac{21D^2}{16}\right]x^2+7x+9$$

$$= \frac{1}{4}\left[x^2-\frac{10x}{4}+\frac{42}{16}\right]+\frac{1}{4}\left[7x-\frac{35}{4}\right]+\frac{9}{4}$$

$$= \frac{1}{4}\left[x^2-\frac{10x}{4}+7x+\frac{42}{16}-\frac{35}{4}+9\right]$$

$$= \frac{1}{4}\left[x^2+\frac{18x}{4}+\frac{46}{16}\right]$$

$$= \frac{1}{32}\left[8x^2+36x+23\right]$$

Solved Example 79 :

Evaluate P.I. of $(D^2 + 2D + 2)y = x$

Solution :

P.I. is $y = \dfrac{1}{D^2+2D+2}x$

$$= \frac{1}{2\left(1+D+\frac{D^2}{2}\right)}x$$

$$= \frac{1}{2}\left(1+D+\frac{D^2}{2}\right)^{-1} x$$

$$= \frac{1}{2}\left[1-D-\frac{D^2}{2}+\left(D+\frac{D^2}{2}\right)^2\right]x$$

$$= \frac{1}{2}\left[1-D-\frac{D^2}{2}+D^2+D^3+\frac{D^4}{4}\right]x$$

$$= \frac{1}{2}[1-D]x$$

….[neglecting higher powers]

$$= \frac{1}{2}[x-1]$$

Solved Example 80 :

Evaluate P.I. of $\left(D^4+D^2\right)y=96x^2$

Solution :

P.I. is $y=\dfrac{1}{D^2\left(D^2+1\right)}96x^2$

$$=96\cdot\frac{1}{D^2}\left(1+D^2\right)^{-1}x^2$$

$$=96\cdot\frac{1}{D^2}\left(1-D^2+D^4-D^6\right)x^2$$

$$=96\left(\frac{1}{D^2}-1+D^2-D^3\right)x^2$$

$$=96\left(\frac{x^4}{12}-x^2+2\right)$$

$$=8x^4-96x^2+192$$

Solved Example 81 :

Evaluate P.I. of $\left(D^3+5D^2+6D\right)y=x^3$

Solution :

P.I. is

$$y=\frac{1}{D^3+5D^2+6D}x^3$$

$$=\frac{1}{6D}\frac{1}{\left[1+\frac{5D}{6}+\frac{D^2}{6}\right]}x^3$$

$$=\frac{1}{6D}\left[1+\frac{5}{6}D\left(1+\frac{D}{5}\right)\right]^{-1}x^3$$

$$=\frac{1}{6D}\left[1-\frac{5}{6}D\left(1+\frac{D}{5}\right)+\frac{25}{36}D^2\left(1+\frac{D}{5}\right)^2\right.$$

$$\left.-\frac{125}{216}D^3\left(1+\frac{D}{5}\right)^3\right]x^3$$

$$=\frac{1}{6D}\left[1-\frac{5}{6}D-\frac{D^2}{6}+\frac{25}{36}D^2\right.$$

$$\left.+\frac{5}{18}D^3-\frac{125}{216}D^3\right]x^3$$

….[Neglecting terms above D^3]

$$=\frac{1}{6D}\left[x^3-\frac{5}{6}\left(3x^2\right)-\frac{1}{6}(6x)+\frac{25}{36}(6x)\right.$$

$$\left.+\frac{5}{18}(6)-\frac{125}{216}(6)\right]$$

$$=\frac{1}{6D}\left[x^3-\frac{5}{2}x^2-x+\frac{25}{6}x+\frac{5}{3}-\frac{125}{36}\right]$$

$$=\frac{1}{6}\left[\frac{x^4}{4}-\frac{5x^3}{6}+\frac{19}{12}x^2-\frac{65}{36}x\right]$$

Solved Example 82 :

Evaluate P.I. of $\left(D^2+9\right)y=\cos^2\left(\dfrac{3x}{2}\right)$

Solution :

P.I. is $y=\dfrac{1}{D^2+9}\cos^2\dfrac{3x}{2}$

Now $\cos^2\dfrac{3x}{2}=\dfrac{1}{2}[1+\cos 3x]$

$$\cos^2\frac{A}{2}=\frac{1}{2}[1+\cos A]$$

$$\therefore\quad y=\frac{1}{2}\frac{1}{D^2+9}1+\frac{1}{2}\frac{1}{D^2+9}\cos 3x$$

$$=\frac{1}{2}\frac{1}{9\left(1+\frac{D^2}{9}\right)}1$$

$$+\frac{1}{2}\frac{(-1)^1x^1}{1!(2\times 3)}\left(\cos 3x+\frac{\pi}{2}\right)$$

$$=\frac{1}{2}\frac{1}{9}\left(1+\frac{D^2}{9}\right)^{-1}+\frac{1}{2}\frac{(-1)}{6}x(-\sin 3x)$$

$$= \frac{1}{2}\frac{1}{9}\left(1-\frac{D^2}{9}\right)+\frac{1}{2}\frac{x\sin 3x}{6}$$

$$= \frac{1}{18}+\frac{x\sin 3x}{12} = \frac{1}{36}\left[2+3x\sin 3x\right]$$

PROBLEMS BASED ON TYPE 4

Solved Example 83 :

Find P.I. of $(D^2-4D+3)y = e^x\cos 2x$

Solution :

P.I. is $y = \left(\frac{1}{D^2-4D+3}\right)e^x\cos 2x$

$$= \left(\frac{e^x}{(D+1)^2-4(D+1)+3}\right)\cos 2x$$

$$= e^x\cdot\frac{1}{D^2-2D}\cos 2x$$

$$= e^x\cdot\frac{1}{-4-2D}\cos 2x$$

$$= e^x\cdot -\frac{1}{2}\cdot\frac{1}{D+2}\cos 2x$$

$$= -\frac{e^x}{2}\cdot\frac{D-2}{D^2-4}\cos 2x$$

$$= -\frac{e^x}{2}\frac{(D-2)}{-4-4}\cos 2x$$

$$= \frac{e^x}{16}(-2\sin 2x-2\cos 2x)$$

$$= -\frac{e^x}{8}(\sin 2x+\cos 2x)$$

Solved Example 84 :

Solve $(D^2-2D+2)y = e^x\sin x$

Solution :

Auxiliary equation is

$D^2-2D+2=0$

$\therefore\quad D = \frac{2\pm\sqrt{4-4.2}}{2}$

$\therefore\quad D = 1\pm i$

C.F. is $y = e^x(c_1\cos x+c_2\sin x)$

P.I. is $y = \left(\frac{1}{D^2-2D+2}\right)e^x\sin x$

$$= e^x\cdot\frac{1}{(D+1)^2-2(D+1)+2}\sin x$$

$$= e^x\cdot\frac{1}{D^2+1}\sin x$$

$$= e^x\cdot\frac{1}{(D+i)(D-i)}\sin x$$

$$\ldots\begin{bmatrix} e^{ix} = \cos x+i\sin x \\ \text{I.P.} = \sin x\end{bmatrix}$$

$= e^x$ Imaginary part of

$$\left[\frac{1}{(D+i)(D-i)}\right]e^{ix}$$

$= e^x$ Imaginary part of

$$\left[\frac{1}{(i+i)}\frac{x^1e^{ix}}{1!}\right]$$

$$\left[\frac{1}{f(D)}e^{ax} = \frac{1}{\phi(a)}\frac{x^re^{ax}}{r!}\right]$$

$= e^x$ Imaginary part of $\left[\frac{1}{2i}xe^{ix}\right]$

$= e^x$ Imaginary part of

$$\left[\frac{i}{2i^2}x(\cos x+i\sin x)\right] \qquad [i^2=-1]$$

$= e^x$ Imaginary part of

$$\left[-\frac{1}{2}ix(\cos x+i\sin x)\right]$$

$$= e^x\left(-\frac{1}{2}x\cos x\right)$$

G.S. is

$$y = e^x\left(C_1\cos x + C_2\sin x\right) - \frac{1}{2}xe^x\cos x$$

Solved Example 85 :

Solve $\left(D^3 - 3D^2 + 3D - 1\right)y = x^2e^x$

Solution :

A.E. is $\left(D^3 - 3D^2 + 3D - 1\right) = 0$

$\therefore \quad (D-1)^3 = 0$

$\therefore \quad D = +1, +1, +1$

$\therefore \quad$ C.F. is

$$y = \left(C_1x^2 + C_2x + C_3\right)e^x$$

P.I. is

$$y = \left(\frac{1}{D^3 - 3D^2 + 3D - 1}\right)x^2e^x$$

$$= e^x\left[\frac{1}{(D+1)^3 - 3(D+1)^2 + 3(D+1) - 1}\right]x^2$$

$$= e^x\frac{1}{D^3}x^2 = \frac{e^xx^5}{60}$$

G.S. is

$$y = \left(C_1x^2 + C_2x + C_3\right)e^x + \frac{e^xx^5}{60}$$

Solved Example 86 :

Evaluate P.I. of

$$\left(D^2 - 4D + 4\right)y = 5x^2e^{2x}\sin 2x$$

Solution :

P.I. is

$$y = \frac{1}{D^2 - 4D + 4}5x^2e^{2x}\sin 2x$$

$$= \frac{1}{(D-2)^2}5x^2e^{2x}\sin 2x$$

$$= 5e^{2x}\frac{1}{(D+2-2)^2}x^2\sin 2x$$

$$= 5\cdot e^{2x}\frac{1}{D^2} \quad \text{imaginary part of } x^2e^{i2x}$$

[e^{i2x} = cos 2x + i sin 2x I.P. = sin2x]

$$= \text{Imaginary part of } 5\cdot e^{2x}e^{i2x}\frac{1}{(D+2i)^2}x^2$$

Now $\left[\frac{1}{(D+2i)}\right]x^2 = -\frac{1}{4}\left[1+\frac{D}{2i}\right]^{-2}x^2$

$$= -\frac{1}{4}\left[1 - \frac{2D}{2i} - \frac{3D^2}{4}\right]x^2$$

$$= -\frac{1}{4}x^2 + \frac{x}{2i} + \frac{3.2}{16}$$

Now P.I.

= Imaginary part of

$$\left[5e^{2x}e^{2ix}\left(-\frac{1}{4}x^2 + \frac{x}{2i} + \frac{3}{8}\right)\right]$$

= Imaginary part of

$$\left[5e^{2x}\left(\cos 2x + i\sin 2x\right)\left(-\frac{1}{4}x^2 - \frac{xi}{2} + \frac{3}{8}\right)\right]$$

$$= 5e^{2x}\left(-\frac{x^2}{4}\sin 2x - \frac{x}{2}\cos 2x + \frac{3}{8}\sin 2x\right)$$

1.12 Euler-Cauchy Equation

The Euler-Cauchy equation is

$$a_0x^ny^n + a_1x^{n-1}y^{n-1} + ... + a_{n-1}xy' + a_ny = r(x), \quad x \neq 0$$

where $a_0, a_1, ...a_n$ are constants

Consider a corresponding homogeneous equation

$$a_0x^2y'' + a_1xy' + ... + a_2y = 0 \qquad \ldots(A)$$

We try to find solution of form $y = x^m$

$\therefore \quad y' = mx^{m-1}$

$y'' = m(m-1)x^{m-2}$

on substituting in equation (A), we get

$$\left[a_0m(m-1) + a_1m + a_2\right]x^m = 0$$

Canceling x^m, we get

A. E. is $a_0m(m-1) + a_1m + a_2 = a_0m^2 + (a_1 - a_0)m + a_2 = 0 \qquad \ldots(B)$

Auxiliary equation has two roots m = m_1m_2 which may real and distinct, real and equal, or complex conjugates. Accordingly the equation will have its general solution.

Real and Distinct Roots

Let m_1 and m_2 be real and distinct, then the two linearly independent solutions are

$$y_1(x) = x^{m_1}$$

$$y_2(x) = x^{m_2}$$

General solution is $y(x) = Ax^{m_1} + Bx^{m_2}$

A and B are arbitrary constants.

Real and Equal Roots

Let the real and equal root be m.

General solution is $y = Ay_1 + By_2 = (A + B\,\ell n\,x)y_1 = x^m(A + B\,\ell n\,x)$

where $\quad m = \dfrac{(a_0 - a_1)}{2a_0}$

Complex Roots

Here $m = p \pm iq$, roots are complex conjugate pair

Two linearly independent solutions are

$$y_1(x) = x^p \cos(q\,\ell n x)$$
$$y_2(x) = x^p \sin(q\,\ell n x)$$

General solution is $y = y_1(x) + y_2(x)$

General solution is $y(x) = x^p\,(A\cos(q\,\ell n\,x) + B\sin(q\,\ell n\,x))$

Case	Roots	General Solution
1. Real and distinct	**m_1 and m_2**	**$y = Ax^{m_1} + Bx^{m_2}$**
2. Real and equal	**m**	**$y = x^m\,(A + B\,\ell n\,x)$**
3. Complex roots	**$m = p \pm iq$**	**$y = x^p\,(A\cos(q\,\ell n\,x) + B\sin(q\,\ell n\,x))$**

Solved Example 87 :

Solve $x^2y'' + 2xy' - 2y = 0$

Solution :

Comparing with

$$a_0x^ny^n + a_1x^{n-1}y^{n-1} + \ldots\ldots + a_{n-1}xy' + a_ny = r(x)$$

A.E. is $a_0m(m-1) + a_1m + a_2$

$$= a_0m^2 + (a_1 - a_0)\,m + a_2$$
$$= 0$$

Here, $a_0 = 1$, $a_1 = 2$ and $a_2 = -2$

$\therefore$ A.E. is $m^2 + m - 2 = 0$

or $(m + 2)(m - 1) = 0$

$\therefore$ $m = 1, -2$

Two linearly independent solutions are

$$y_1(x) = x$$
$$y_2(x) = x^{-2}$$

G.S. is $y(x) = Ax + B/x^2$

Solved Example 88 :

Solve $2x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} - 6y = 0$

Solution :

Comparing with

$$a_0x^ny^n + a_1x^{n-1}y^{n-1} + \ldots\ldots + a_{n-1}xy' + a_ny = r(x)$$

Here, $a_0 = 2$, $a_1 = 1$ and $a_2 = -6$

A.E. is $a_0m(m-1) + a_1m + a_2$

$$= a_0m^2 + (a_1 - a_0)\,m + a_2$$
$$= 0$$

$\therefore$ $2m^2 - m - 6 = 0$

or $(m - 2)(2m + 3) = 0$

$\therefore$ $m = 2, -3/2$

Two linearly independent solutions are

$$y_1(x) = x^2$$
$$y_2(x) = x^{-3/2}$$

G.S. is $y(x) = Ax^2 + \dfrac{B}{x\sqrt{x}}$

Solved Example 89 :

Solve $4x^2 \frac{d^2y}{dx^2} + y = 0$

Solution :

Here, $a_0 = 4$, $a_1 = 0$, $a_2 = 1$

A. E.

$a_0 m(m-1) + a_1 m + a_2 = 4m^2 - 4m + 1 = 0$

or $(2m-1)^2 = 0$

$m = \frac{1}{2}, \frac{1}{2}$

G.S. is $y(x) = (A + B\,\ell n\,x)x^{1/2} \quad x > 0$

Solved Example 90 :

Solve $4x^2 \frac{d^2y}{dx^2} + 8x\frac{dy}{dx} + 17y = 0$

Solution :

Here $a_0 = 4$, $a_1 = 8$, and $a_2 = 17$

A. E.

$a_0 m(m-1) + a_1 m + a_2 = 4m^2 + 4m + 17 = 0$

$m = \frac{-4 \pm \sqrt{16-272}}{8}$

$= \frac{-4 \pm 16i}{8} = -\frac{1}{2} \pm 2i = p \pm iq$

G.S. is

$y(x) = Ax^{-1/2}\cos(2\ell n\,x) + B\,x^{-1/2}\sin(2\ell n\,x)$

Euler-Cauchy Non Homogenous Equations

Consider second order non–homogenous Euler-Cauchy equation

$$a_0 x^2 y'' + a_1 xy' + a_2 y = r(x) \quad a_0 \neq 0, \ x \neq 0$$

put $x = e^z$

$$\mathbf{x^n D^n y = \theta(\theta-1)(\theta-2).....[\theta-(n-1)]y}$$

We get a linear Euler-Cauchy equation with constant coefficients which can be easily solved.

Solved Example 91 :

Solve $x^2 \frac{d^2y}{dx^2} - 3x\frac{dy}{dx} - 5y = \sin(\log x)$

Solution :

Put $x = e^z$ and $\theta = \frac{d}{dz}$

or $z = \log x$

$\therefore$ The given equation reduces to

$[\theta(\theta - 1) - 3\theta - 5]y = \sin z$

$\ldots.\left[x^n D^n y = \theta(\theta-1)(\theta-2).....[\theta-(n-1)]y\right]$

$\therefore \quad (\theta^z - 4\theta - 5)y = \sin z$

A.E. is $m^2 - 4m - 5 = 0$

$\therefore \quad (m-5)(m+1) = 0$

$\therefore \quad m = 5, -1$

C.F. is

$y = C_1 e^{5z} + C_2 e^{-z}$

$= C_1 e^{5\log x} + C_2 e^{-\log x} \quad = C_1 x^5 + C_2 x^{-1}$

P.I. is

$$y = \left[\frac{1}{\theta^2 - 4\theta - 5}\right]\sin z$$

$$= \left[\frac{1}{-1 - 4\theta - 5}\right]\sin z$$

$$= \left[\frac{1}{-2(2\theta + 3)}\right]\sin z$$

$$= \left[\frac{2\theta - 3}{-2(4\theta^2 - 9)}\right]\sin z$$

$$= \left[\frac{2\theta - 3}{26}\right]\sin z$$

$$= \frac{2\cos z - 3\sin z}{26}$$

$$= \frac{1}{13}\cos(\log x) - \frac{3}{26}\sin(\log x)$$

G.S. is

$$y = C_1 x^5 + C_2 x^{-1} + \frac{1}{13}\cos(\log x) - \frac{3}{26}\sin(\log x)$$

Solved Example 92 :

Solve

$$x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} + 4y = \cos(\log x) + x\sin(\log x)$$

Solution :

Put $z = \log x$ and $\theta = \frac{d}{dz}$

The equation reduces to

$[\theta(\theta - 1) - \theta + 4]y = \cos z + e^z \sin z$

A.E. is $m^2 - 2m + 4 = 0$

$\therefore \quad m = 1 \pm i\sqrt{3}$

Hence C.F. is

$$y = e^z(C_1\cos\sqrt{3}z + C_2\sin\sqrt{3}z)$$

$$= x\left[C_1\cos\left(\sqrt{3}\log x\right) + C_2\sin\left(\sqrt{3}\log x\right)\right]$$

P.I. is

$$y = \frac{1}{\theta^2 - 2\theta + 4}\cos z + \frac{1}{\theta^2 - 2\theta + 4}e^x\sin z$$

$$= \frac{1}{3 - 2\theta}\cos z + e^z \cdot \frac{1}{(\theta + 1)^2 - 2(\theta + 1) + 4}\sin z$$

$$= \frac{3 + 2\theta}{9 - 4\theta^2}\cos z + e^z\frac{1}{-1 + 3}\sin z$$

$$= \frac{3 + 2\theta}{13}\cos z + \frac{1}{2}e^z\sin z$$

$$= \frac{1}{13}[3\cos z - 2\sin z] + \frac{1}{2}e^z\sin z$$

$$= \frac{1}{13}[3\cos(\log x) - 2\sin(\log x)] + \frac{1}{2}x\sin(\log x)$$

G.S. is

$$y = x\left[C_1\cos\left(\sqrt{3}\log x\right) + C_2\sin\left(\sqrt{3}\log x\right)\right]$$

$$+ \frac{1}{13}[3\cos(\log x) - 2\sin(\log x)]$$

$$+ \frac{1}{2}x\sin(\log x)$$

Solved Example 93 :

Solve $x^2\frac{d^2y}{dx^2} + 4x\frac{dy}{dx} + 2y = x\log x$

Solution :

Put $z = \log x$ and $\theta = \frac{d}{dz}$

Equation reduces to

$[\theta(\theta - 1) + 4\theta + 2]\, y = ze^z$

i.e. $(\theta^2 + 3\theta + 2)y = ze^z$

$\therefore$ A.E. is $m^2 + 3m + 2 = 0$

$\therefore \quad (m + 1)(m + 2) = 0$

$\therefore \quad m = -1, -2$

C.F. is $y = C_1e^{-z} + C_1e^{-2z}$

$$= C_1x^{-1} + C_2x^{-2}$$

P.I. is $y = \dfrac{1}{\theta^2 + 3\theta + 2} ze^z$

$$= e^z \frac{1}{(\theta+1)^2 + 3(\theta+1) + 2} z$$

$$= e^z \frac{1}{\theta^2 + 5\theta + 6} z$$

$$= \frac{e^z}{6}\left[1 + \frac{5\theta + \theta^2}{6}\right]^{-1} z$$

$$= \frac{e^z}{6}\left[1 - \frac{5\theta}{6}\right] z = \frac{e^z}{6}\left[z - \frac{5}{6}\right]$$

$$= \frac{x}{6}\left[\log x - \frac{5}{6}\right]$$

G.S. $y = C_1x^{-1} + C_2x^{-2} + \dfrac{x}{6}\left[\log x - \dfrac{5}{6}\right]$

Solved Example 94 :

Solve $(2x+1)^2 y'' - 2(2x+1)y^1 - 12y = 6x$

Solution :

Put 2x + 1 = z

$\therefore \quad \dfrac{dz}{dx} = 2 \quad$ and $\quad x = \dfrac{z-1}{z}$

Now $\dfrac{dy}{dx} = \dfrac{dy}{dz}\dfrac{dz}{dx} = 2\dfrac{dy}{dz}$

and $\dfrac{d^2y}{dx^2} = 2\dfrac{d^2y}{dz^2}\dfrac{dz}{dx} = 4\dfrac{d^2y}{dz^2}$

Given equation reduces to a linear homogeneous equation

$$\therefore 4z^2\frac{d^2y}{dz^2} - 4z\frac{dy}{dz} - 12y = 6\left(\frac{z-1}{2}\right)$$

Now, put $u = \log z$ and $\theta = \dfrac{d}{du}$

substituting in (1) we get

$$[4\theta(\theta-1) - 4\theta - 12]y = 3(e^u - 1)$$

$$\therefore (4\theta^2 - 8\theta - 12)y = 3(e^u - 1)$$

$$\therefore (\theta^2 - 2\theta - 3)y = \frac{3}{4}(e^u - 1)$$

A.E is $m^2 - 2m - 3 = 0$

$$\therefore (m-3)(m+1) = 0$$

$$\therefore m = 3, -1$$

C.F is

$$y = C_1^{3u} + C_2e^{-u}$$

$$= C_1e^{3\log z} + C_2e^{-\log z}$$

$$= C_1z^3 + C_2z^{-1}$$

$$= C_1(2x+1)^3 + C_2(2x+1)^{-1}$$

P.I. is

$$y = \frac{3}{4}\left[\frac{1}{\theta^2 - 2\theta - 3}\right](e^u - 1)$$

$$= \frac{3}{4}\left(\frac{e^u}{-4} + \frac{e^0}{3}\right) \qquad \left[\begin{array}{l} \dfrac{e^u}{1^2 - 2(1) - 3} = \dfrac{e^u}{-4} \\ \dfrac{-e^0}{0 - 2(0) - 3} = \dfrac{1}{3} \end{array}\right]$$

$$= \frac{3}{4}\left(\frac{-z}{4} + \frac{1}{3}\right) = \frac{3}{4}\left(\frac{2x+1}{-4} + \frac{1}{3}\right) = \frac{-3x}{8} + \frac{1}{16}$$

F.S is

$$y = C_1(2x+1)^3 + C_2(2x+1)^{-1} - \frac{3x}{8} + \frac{1}{16}$$

Solved Example 95 :

Solve $x^2y'' + 3xy' + y = \dfrac{1}{(1-x)^2}$

Solution :

put $z = \log x$ and $\theta = \dfrac{d}{dz}$

Equation reduces to

$$\left[\theta(\theta-1)+3\theta+1\right]y = \frac{1}{(1-x)^2}$$

$$\therefore \quad \left(\theta^2+2\theta+1\right)y = \frac{1}{(1-x)^2}$$

A.E. is $m^2+2m+1=0$

$\therefore \quad (m+1)^2 = 0$

$\therefore \quad m = -1, -1$

C.F. is

$y = e^{-z}\left(C_1 + C_2 z\right)$

$= x^{-1}\left(C_1 + C_2 \log x\right)$

P.I. is

$$y = \left[\frac{1}{(\theta+1)^2}\right]\frac{1}{(1-x)^2}$$

$$= \frac{1}{(\theta+1)}\left[x^{-1}\int\frac{dx}{(1-x)^2}\right]$$

$$= \frac{1}{(\theta+1)}\left(\frac{1}{x(1-x)}\right)$$

$$= x^{-1}\int\frac{1}{x(1-x)}dx$$

$$= x^{-1}\int\frac{1}{x}+\frac{1}{1-x}dx = \frac{1}{x}\left[\log x - \log(1-x)\right]$$

$$= \frac{1}{x}\log\left(\frac{x}{1-x}\right)$$

G.S. is $y = x^{-1}\left(C_1 + C_2\log x\right) + \dfrac{1}{x}\log\left(\dfrac{x}{1-x}\right)$

1.13 Method of Variation of Parameters

For solving linear differential equation

Consider linear differential equation

$$y' + f(x)y = r(x) \qquad .....(1)$$

We know that solution of corresponding homogenous equation $y'+f(x)y = 0$ is

$$v(x) = e^{-\int f(x)dx} \qquad ....(2)$$

using this function v(x), we will determine function u(x) such that

$$y(x) = u(x).v(x) \qquad ....(3)$$

This can be generalized to equation of higher order.

For solving non–homogeneous equation

Consider a non–homogeneous equation

$$a(x).\frac{d^2y}{dx^2} + b(x).\frac{dy}{dx} + c(x)y = f(x) \qquad ...(1)$$

and let $y_1(x)$ and $y_2(x)$ be two linearly independent solution of equation:

$$a(x)\frac{d^2y}{dx^2}+b(x)\frac{dy}{dx}+c(x)y=0 \qquad ...(2)$$

$\therefore$ The general solution of equation (2) is

$y = c_1\, y_1(x) + c_2\, y_2\,(x)$.

we now try to find solution of equation (1) by replacing constants c_1 and c_2 by function of x.

$\therefore \quad y = v_1\,(x)\,.\,y_1\,(x)\,. + v_2\,(x)\,.\,y_2\,(x) \qquad ...(3)$

$v_1\,(x)$ and $v_2\,(x)$ can be found out by substituting equation (3) in (1).

and imposing a condition that

$$v_1'y_1+v_2'y_2=0 \qquad ...(4)$$

using this condition and calculating y'' , we get another condition

$$a(v_1'y_1'+v_2'y_2')=f(x) \qquad ...(5)$$

from (4) and (5), we can find v_1 and v_2 and thus y.

Solved Example 96 :

$y' - \frac{2y}{x} = x^2 \cos 3x$

Solution :

$y' - \frac{2y}{x} = x^2 \cos 3x \qquad ...(1)$

Now, homogeneous equation $y' - \frac{2y}{x} = 0$.

It's solution is x^2.

$\therefore$ Let $y = ux^2 \qquad ...(2)$

$\Rightarrow \quad y' = u'x^2 + 2ux \qquad ..(3)$

Substituting (3) in (1), we get

$u'x^2 + 2ux - 2ux = x^2 \cos 3x$

$\Rightarrow \quad u' = \cos 3x.$

$\Rightarrow \quad u = \frac{1}{3}\sin 3x + c$

$\therefore \quad y = \left(\frac{1}{3}\sin 3x + c\right)x^2$

..... is required solution.

Solved Example 97 :

Solve $\frac{d^2y}{dx^2} + y = \sec x$

Solution :

The two linear independent solution of homogeneous equation

$\frac{d^2y}{dx^2} + y = 0$ are $y_1 = \cos x$, $y_2 = \sin x$

$\therefore$ The general solution is

$y = v_1 \cos x + v_2 \sin x$

... [v_1 and v_2 are function of x]

$\therefore \quad y' = -v_1 \sin x + v_2 \cos x + v_1' \cos x$

$+ \; v_2' \sin x$

Imposing condition

$v' \cos x + v_2' \sin x = 0 \qquad ...(1)$

$\therefore \quad y' = -v_1 \sin x + v_2 \cos x$

$\therefore \quad y'' = -v_1 \cos x - v_2 \sin x -$

$v_1' \sin x + v_2' \cos x \qquad ...(2)$

Substituting in given D.E., we get.

$$-v_1' \sin x + v_2' \cos x = \sec x \qquad \ldots(3)$$

Solving (1) and (3), we get

$$v_1' = -\sin x.\sec x, \quad v_2' = \cos x.\sec x$$

$$= -\tan x \quad , \quad = 1$$

$$\Rightarrow \quad v_1 = -\int \tan x\, dx = -\ln \sec x + A$$

$$v_2 = \int dx = x + B$$

Hence

$$y = (-\ln \sec x + A) \cos x + (x + B) \sin x$$

… required solution

Solved Example 98 :

Apply the method of variation of parameters to solve.

$$y'' - y = \frac{2}{(1+e^x)}$$

Solution :

The given equation in standard form is

$$y'' - y = \frac{2}{(1+e^x)} \qquad \ldots(1)$$

Consider $y'' - y = 0$ or $(D^2 - 1)\, y = 0$ ….(2)

Auxiliary equation of (2) is $D^2 - 1 = 0$.

So $D = \pm 1$ and hence solution of (2) is

$y = ae^x + be^{-x}$ so that $ae^x + be^{-x}$ is C.F. of (1), a and b being arbitrary constants.

Let $y = Ae^x + Be^{-x}$ ….(3)

be the general solution of (1). Then A and B are functions of x to be determined.

Choosing A and B such that

$$A'e^x + B'e^{-x} = 0 \qquad \ldots(4)$$

Differentiating (3),

$$y' = A'e^x + Ae^x + B'e^{-x} - Be^{-x}$$

or $y' = Ae^x - Be^{-x}$, [using (4)] ….(5)

Differentiating (5),

$$y'' = A'e^x + Ae^x - B'e^{-x} + Be^{-x} \qquad \ldots(6)$$

Using (3) and (6), (1) reduces to

$$A'e^x - B'e^{-x} = \frac{2}{(1+e^x)} \qquad \ldots(7)$$

Adding (4) and (7), we get

$$2A'e^x = \frac{2}{(1+e^x)} \quad \text{or} \quad A' = \frac{1}{[e^x(1+e^x)]} \quad \ldots(8)$$

Integrating (8),

$$A = \int \frac{dx}{e^x(1+e^x)} + c_1$$

$$= \int \frac{dz}{z^2(1+z)} + c_1 \quad [\text{Put } e^x = z \therefore dx = dz/z]$$

$$= \int \left[\frac{1}{z^2} - \frac{1}{z} + \frac{1}{1+z}\right] dz + c_1$$

$$= -(1/z) - \log z + \log(1+z) + c_1$$

$$= c_1 - e^{-x} + \log[(1+e^x)/e^x] \qquad (\because z = e^x)$$

$$A = c_1 - e^{-x} + \log(1+e^{-x}) \qquad \ldots(9)$$

Using (8), (4) gives

$$B' = -\frac{e^x}{e^x(1+e^x)e^{-x}} = -\frac{e^x}{1+e^x}$$

Integrating,

$$B = \int \frac{-e^x}{1+e^x} dx$$

$$= -\log(1+e^x) + c_2 \qquad \ldots(10)$$

Using (9) and (10) in (3), the required solution is

$$y = [c_1 - e^{-x} + \log(1+e^{-x})]\, e^x + [c_2 - \log(1+e^x)]\, e^{-x}$$

or $y = c_1e^x + c_2e^{-x} - 1 + e^x \log(1+e^{-x}) - e^{-x}\log(1+e^x)$

Solved Example 99 :

Apply the method of variation of parameters to solve

$$x^2y'' + xy' - y = x^2e^x$$

Solution :

$x^2y'' + xy' - y = x^2e^x$ ....(1)

Consider $x^2y'' + xy' - y = 0$ ....(2)

Put $x = e^z$ and $\theta = d/dz$,

(2) becomes

$[\theta(\theta - 1) + \theta - 1]\, y = 0$ or $(\theta_1^2 - 1)\, y = 0$

Here auxiliary equation is $\theta_1^2 - 1 = 0$

so that $\theta_1 = \pm 1$

So solution of (2) is

$y = ae^z + be^{-z} = ax + bx^{-1}$ $(\because x = e^x)$

Hence the C.F. of (1) is $ax + bx^{-1}$, a and b being arbitrary constants.

Let $y = Ax + Bx^{-1}$ ....(3)

be the general solution of (1). Then A and B are functions of x to be determined. Choosing A and B such that

$A'x + B'x^{-1} = 0$ ....(4)

Differentiating (3),

$y' = A'x + A + B'x^{-1} - Bx^{-2}$

or $y' = A - Bx^{-2}$ [using (4)] ....(5)

Differentiating (5),

$y'' = A' - B'x^2 + 2Bx^{-3}$ ....(6)

Using (3), (5) and (6), (1) reduces to

$A' - B'x^{-2} = e^x$ ....(7)

Multiplying (7) by x and adding to (4), we get

$2A'x = xe^x$ or $A' = \frac{1}{2}e^x$ ....(8)

Integrating (8), $A = c_1 + \frac{1}{2}e^x$

Using (8), (4) gives

$B' = -\frac{1}{2}x^2e^x$ .....(9)

Integrating and using chain rule of integration by parts,

$$B = -\frac{1}{2}\left[(x^2)(e^x) - (2x)(e^x) + (2)(e^x)\right] + c_2$$

$$= c_2 - \frac{1}{2}e^x(x^2 - 2x + 2) \quad ....(10)$$

Using (9) and (10) in (3), the required solution is

$$y = \left(c_1 + \frac{1}{2}e^x\right)x + \left[c_2 - \frac{1}{2}e^x(x^2 - 2x + 2)\right]x^{-1}$$

$$= c_1x + c_2x^{-1} + e^x(1 - x^{-1})$$

Partial Differential Equations

Partial differential equations arise when the functions involved depend on two or more independent variables. These variables may be time or one of the co-ordinate space.

For Example :

(i) $x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=0$ (ii) $x^2\frac{\partial^2 z}{\partial x^2}+y^2\frac{\partial^2 z}{\partial y^2}=z^2$ (iii) $\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2}=0$

When x and y are independent variables and z is dependent variable, we use following notation.

$\frac{\partial z}{\partial x}=p,\quad \frac{\partial z}{\partial y}=q,\quad \frac{\partial^2 z}{\partial x^2}=r,\quad \frac{\partial^2 z}{\partial x\partial y}=s,\quad \frac{\partial^2 z}{\partial y^2}=t$

The order of a partial differential equation is same as the order of the highest differential co-efficient occurring in it.

For Example :

$\frac{\partial^3 w}{\partial x^3}=u\frac{\partial u}{\partial y}$ is a partial differential equation of order 3 and degree 1.

1.14 Formation of Partial Differential Equation

Method of Elimination of Arbitrary Constants

For Example :

Form partial differential equation from

$x^2+y^2+(z-c)^2=a^2$ …[where a, c are arbitrary constant] …(1)

differentiating partially (1) w.r.t. 'x' , we get

$2x+2(z-c)\frac{\partial z}{\partial x}=0$ or $x+(z-c)\,p=0$ …(2)

differentiating (1) partially w.r.t. 'y', we get.

$2y+2(z-c)\frac{\partial z}{\partial y}=0$ or $y+(z-c)\,q=0$ …(3)

eliminating 'c', we get

$\frac{x}{p}=\frac{y}{q}$ or $xq-py=0$

$\therefore$ Partial differential equation is

$$x\frac{\partial z}{\partial y} - y\frac{\partial z}{\partial x} = 0 \quad \text{or} \quad xq - yp = 0$$

<u>Note :</u> If number of constants to be eliminated is just equal to number of independent variable, then we will get a partial differential equation of first order. But if numbers of constants are greater than number of independent variables, then partial D.E. will be of higher order].

Formation of PDE by eliminating of arbitrary functions

Consider z = f (x + ay), where 'a' is a constant but function is arbitrary

let $\quad r = x + ay \quad \Rightarrow z = f(r)$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial r}.\frac{\partial r}{\partial x} = \frac{\partial z}{\partial r}.(1) = f'(r) = f'(x + ay) \qquad \ldots(1)$$

Similarly

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial r}.\frac{\partial r}{\partial y} = af'(r) = af'(x + ay) \qquad \ldots(2)$$

eliminating f′ (x + ay) , we get

$$\frac{\partial z}{\partial y} = a\frac{\partial z}{\partial x} \qquad \ldots\text{required PDE.}$$

<u>For Example</u> :

(1) form PDE from $z = f(x^2 - y^2)$

$$\therefore \quad P = \frac{\partial z}{\partial x} = f'(x^2 - y^2)(2x)$$

$$q = \frac{\partial z}{\partial y} = f'(x^2 - y^2)(-2y)$$

$\therefore$ eliminating $f'(x^2 - y^2)$, we get

$$\frac{p}{x} + \frac{q}{y} = 0 \qquad \text{or} \qquad py + qx = 0 \ \ldots \text{[required PDE]}$$

1.15 Solution of Partial Differential Equation

There are two distinct types of solution of PDE, one type of solution containing arbitrary constants and the other type containing arbitrary functions.

i.e. for $z = ax + by$ and $z = y\, f\left(\frac{y}{x}\right)$.

are Solution of $px + qy = z$

Note : **(i) A solution which contains maximum arbitrary constants is called complete integral**

(ii) A solution obtained by giving particular values to the arbitrary constants is called a particular integral (solution)]

(iii) A solution which contains the maximum possible number of arbitrary functions is called general integral (solution.)]

To Find General Solution :

(i) let $F(x, y, z, p, q) = 0$ be PDE whose complete solution is

$$\phi(x, y, z, a, b) = 0 \qquad \ldots(1)$$

If one of the two arbitrary constant say $b = f(a)$

then (1) becomes

$$\phi(x, y, z, a, f(a)) = 0 \qquad \ldots(2)$$

differentiating (1), w.r.t. 'a', we get

$$\frac{\partial\phi}{\partial a} + \frac{\partial\phi}{\partial b} f'(a) = 0 \qquad \ldots(3)$$

The elimination of 'a' between (2) and (3), if it can be found out is called G.S. (general solution)

(ii) Another type of solution is called singular integral differentiating partially, (1), w.r.t. 'a' and 'b', we get.

$$\frac{\partial\phi}{\partial a} = 0 \quad \text{and} \quad \frac{\partial\phi}{\partial b} = 0 \qquad \ldots(4)$$

The elimination of 'a' and 'b' from (1) and (4), if it exists is called singular solution.

e.g. Solve $\frac{\partial z}{\partial y} = 0$

$\therefore$ Integrating w.r.t. y, we get z = f (x)

$\therefore$ z = f (x) … [required solution]

Solution of First order PDE

Type I :

Equation of form F (p,q) = 0, where all 3 variables do not occur explicitly.

then let z = ax + by + c be solution

then $p = \frac{\partial z}{\partial x} = a, \qquad q = \frac{\partial z}{\partial y} = b$

on substituting in F (p, q) = 0, we get F (a, b) = 0

i.e. z = ax + by + c will be solution of F (p, q) = 0 if F (a, b) = 0

If we take b = f (a), then

z = ax + f(a) y + c, where 'a' and 'c' are arbitrary constant is the complete solution.

To find general solution put c = ϕ (a), where ϕ is arbitrary function.

Then z = ax + yf (a) + ϕ (a) …(3)

$\therefore$ differentiating w.r.t. + 'a'

0 = x + yf′ (a) + ϕ′ (a) …(4)

The eliminant of 'a' between (3) and (4) is G.S.

Type II : Clairaut's Type

Equation of type z = px + qy + f(p,q) are called clairaut's type.

If z = ax + by + c is solution of given PDE, then c = f (a, b)

$\therefore$ z = ax + by + f (a, b) is the solution and also complete solution

To find singular solution

z = ax + by + f (a, b) …(1)

differentiating partially w.r.t. 'a' and 'b' and equate it to zero.

$\therefore \quad \frac{\partial z}{\partial a} = x + \frac{\partial f}{\partial a} = 0$ …(2)

$\frac{\partial z}{\partial b} = y + \frac{\partial f}{\partial b} = 0$ …(3)

The eliminant of 'a' and 'b' from 1, 2, 3 give singular solution.

Solved Example 100 :

Solve $p^2 + q^2 = 1$

Solution :

let $z = ax + by + c$ be its solution

then $p = a$ and $q = b$

$\therefore a^2 + b^2 = 1$ or $b = \sqrt{1-a^2}$

$\therefore z = ax + y\sqrt{1-a^2} + c$, …required solution

Solved Example 101 :

Solve $z = px + qy - 2\sqrt{pq}$

Solution :

$z = ax + by - 2\sqrt{ab}$

… [complete solution] …(1)

Differentiating (1), partially w.r.t. 'a' and 'b' and equating to zero.

$$x - 2\sqrt{b} \times \frac{1}{2\sqrt{a}} = 0 \qquad \Rightarrow x = \sqrt{\frac{b}{a}}$$

and $$y - 2\sqrt{a} \times \frac{1}{2\sqrt{b}} = 0 \quad \Rightarrow \quad y = \sqrt{\frac{a}{b}}$$

hence $a = \frac{y}{x}$ and $b = \frac{x}{y}$, eliminating 'a' and 'b' from (1) we get.

$z = x + y - 2$ … [singular solution]

<u>Note :</u> The solution f(x, y, z, a, b) = 0 of a first order partial differential equation which contains two arbitrary constants is called complete solution or complete integral.

A solution obtained from the complete integral by giving particular values to the arbitrary constant is called a particular solution or particular integral.

A partial differential equation may have a large number of entirely different solutions.

<u>For Example</u> :

$\mu = x^2 - y^2$

$\mu = \log(x^2 + y^2)$

$\mu = \sin kx \cosh kx$

are the solutions of the Laplace equation

$$\frac{\partial^2 \mu}{\partial x^2} + \frac{\partial^2 \mu}{\partial y^2} = 0$$

The unique solution of a partial differential equation corresponding to the physical problem satisfy certain other conditions at the boundary of the region R. These are known as boundary conditions.

If these conditions are given for the time t = 0, they are known as initial conditions.

1.16 Classification of the Differential Equation of Second Order

Consider the differential equation of second order in two independent variables x and y as

$$A\frac{\partial^2\mu}{\partial x^2}+B\frac{\partial^2\mu}{\partial x\,\partial y}+C\frac{\partial^2\mu}{\partial y^2}+f\left(x, y, \mu, \frac{\partial\mu}{\partial x}, \frac{\partial\mu}{\partial y}\right)=0$$

where, A, B and C are constants or continuous functions of x and y processing continuous partial derivatives and A is positive.

Now the above equation is

elliptic, if $B^2 - 4AC < 0$

hyperbolic, if $B^2 - 4AC > 0$

parabolic, if $B^2 - 4AC = 0$

For Example :

one dimensional wave equation $z_{xx} = z_{tt}$ is a hyperbolic equation.

One dimensional heat flow equation $\mu_t = \mu_{xx}$ is parabolic type. Two dimensional heat flow equation is given by $z_{xx} + z_{yy} = 0$ is elliptic.

1.17 Wave Equation

One-dimensional wave equation is $\frac{\partial^2\mu}{\partial t^2}=c^2\frac{\partial^2\mu}{\partial t^2}$ (equation of the vibration of a string).

Two-dimensional wave equation is $\frac{\partial^2\mu}{\partial t^2}=c^2\left(\frac{\partial^2\mu}{\partial x^2}+\frac{\partial^2\mu}{\partial y^2}\right)$ (equation of the vibration of a rectangular membrance).

Three-dimensional wave equation is $\frac{\partial^2\mu}{\partial t^2}=c^2\left(\frac{\partial^2\mu}{\partial x^2}+\frac{\partial^2\mu}{\partial y^2}+\frac{\partial^2\mu}{\partial z^2}\right)$

Solution of Wave Equation

Wave equation is

$$\frac{\partial^2 y}{\partial t^2}=c^2\frac{\partial^2 y}{\partial x^2} \qquad \text{....(1)}$$

Let its solution be

$$y(x, t) = X(x)\,T(t) \qquad \text{....(2)}$$

Substituting from Eq. (2) in Eq. (1), we get

$$X\frac{d^2T}{dt^2} = c^2T\frac{d^2X}{dx^2}$$

$$\Rightarrow \quad \frac{1}{T}\cdot\frac{d^2T}{dt^2} = \frac{c^2}{X}\cdot\frac{d^2X}{dx^2} \qquad \text{....(3)}$$

Since, x and t are independent variables and left side of Eq. (3) is a function of t alone while right side is a function of x, Eq. (3) will be true only when each side is equal to a constant :

- When constant is negative ($-k^2$)

$$\frac{c^2}{X}\frac{d^2X}{dx^2} = -k^2$$

$$\therefore \quad \frac{d^2X}{dx^2} + \frac{k^2X}{c^2} = 0$$

$$\therefore \quad \left(D^2 + \frac{k^2}{c^2}\right)X = 0$$

A.E. is $D^2 + \frac{k^2}{c^2} = 0$

i.e., $D^2 = \frac{-k^2}{c^2}$

$$\therefore \quad D = \pm\frac{k}{c}i$$

$\therefore$ Solution is $X = A\cos\frac{kx}{c} + B\sin\frac{kx}{c}$

Also, $\frac{1}{T}\frac{d^2T}{dt^2} = -k^2$

$$\therefore \quad \frac{d^2T}{dt^2} + k^2T = 0$$

$\therefore \quad (D^2 + k^2)\,T = 0$

A.E. is $D^2 + k^2 = 0$

i.e. $D^2 = -k^2$

$\therefore \quad D = \pm ki$

$\therefore \quad$ Solution is $T = C\cos kt + D\sin kt$

$\therefore$ Solution for y :

y(x, t) = X(x) T(t)

$$\mathbf{y(x, t) = \left(A\cos\frac{kx}{c} + B\sin\frac{kx}{c}\right)(C\cos kt + D\sin kt)}$$

- When constant is zero

$$\frac{c^2}{X}\frac{d^2X}{dx^2} = 0$$

$$\therefore \quad \frac{d^2X}{dx^2} = 0$$

$$\therefore \quad \frac{dX}{dx} = A$$

X = Ax + B

Similarly, T = (Ct + D)

Solution for y = y(x, t) = X(x) T(t)

$$\mathbf{y(x, t) = (Ax + B)\,(Ct + D)}$$

- When constant is positive (k^2)

$$\frac{c^2}{X}\frac{d^2X}{dx^2} = k^2$$

$$\therefore \quad \frac{d^2X}{dx^2} - \frac{k^2}{c^2}X = 0$$

$$\therefore \quad \left(D^2 - \frac{k^2}{c^2}\right)X = 0$$

A.E. is $D^2 - \frac{k^2}{c^2} = 0$

$$D^2 = \frac{k^2}{c^2}$$

$$D = \pm\frac{k}{c}$$

$\therefore$ Solution is $X = Ae^{-\frac{kx}{c}} + Be^{\frac{kx}{c}}$

Also, $\frac{1}{T}\frac{d^2T}{dt^2} = k^2$

$$\therefore \quad \frac{d^2T}{dt^2} - k^2T = 0$$

$\therefore \quad (D^2 - k^2)\, T = 0$

A.E. is $D^2 - k^2 = 0$

$\therefore \qquad D^2 = k^2$

$D = \pm k$

Solution is $T = Ce^{-kt} + De^{kt}$

Solution for y :

$y(x, t) = X(x)\, T(t)$

$$\mathbf{y(x, t) = \left(A e^{-\frac{kx}{c}} + B e^{\frac{kx}{c}}\right)\left(C e^{-kt} + De^{-kt}\right)}$$

1.18 Heat Equation

One-dimensional heat equation is $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$, where $c^2 = \frac{k}{S\rho}$ is known as diffusivity of the material of the bar.

Solution of the Heat Equation

The heat equation is $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ ....(1)

Let $u(x, t) = X(x)\, T(t)$ ....(2)

where, X is a function of x only and T is a function of t only be a solution of Eq. (1).

Then, $\frac{\partial u}{\partial t} = XT'$ and $\frac{\partial^2 u}{\partial x^2} = X''T$

Substituting in Eq. (1),

$$XT' = c^2 X'' T$$

$$\frac{X''}{X} = \frac{1}{c^2} \cdot \frac{T'}{T} \qquad ....(3)$$

Now, LHS of Eq. (3) is a function of x only and the RHS is a function of t only. Since, x and t are independent variable, this equation can hold only when each side is equal to a constant

- When constant is negative $(-k^2)$

$$\frac{1}{X}\frac{d^2X}{dx^2} = -k^2$$

$$\therefore \quad \frac{d^2X}{dx^2} + k^2X = 0$$

$$(D^2 + k^2)\, X = 0$$

A.E. is $D^2 + k^2 = 0$

$$D^2 = -k^2$$

$$D = \pm k\, i$$

Solution is $X = A \cos kx + B \sin kx$

Also, $\frac{1}{c^2T}\frac{dT}{dt} = -k^2$

$$\therefore \quad \frac{dT}{T} = -k^2c^2\, dt$$

$$\therefore \quad \ell n\, T = -k^2c^2t + \ell n\, C$$

$$T = Ce^{(-k^2c^2t)}$$

Solution for u :

$u(x, t) = X(x)\, T(t)$

$$\mathbf{u(x, t) = (A \cos kx + B \sin kx)\,(C\, e^{-k^2c^2t})}$$

- When constant is zero

$$\frac{1}{X}\frac{d^2X}{dx^2} = 0$$

$$\therefore \quad \frac{d^2X}{dx^2} = 0$$

$$\frac{dX}{dx} = A$$

$$X = Ax + B$$

Also, $\frac{1}{c^2T}\frac{dT}{dt} = 0$

$$\frac{dT}{dt} = 0$$

$$T = C$$

Solution for u :

$u(x, t) = (Ax + B)\, C$

$$\mathbf{u(x, t) = Dx + E}$$

- When constant is positive (k^2)

$$\frac{1}{X}\frac{d^2X}{dx^2} = k^2$$

$$\therefore \quad \frac{d^2X}{dx^2} - k^2X = 0$$

$$(D^2 - k^2) = 0$$

A.E. is $D^2 - k^2 = 0$

$$D^2 = k^2$$

$$D = \pm k$$

Solution is $X = A\,e^{-kx} + B\,e^{kx}$

Also, $\frac{1}{c^2T}\frac{dT}{dt} = k^2$

$$\therefore \quad \frac{dT}{T} = k^2c^2dt$$

$$\therefore \quad \ell n\,T = k^2c^2t + \ell n\,C$$

$$T = C\,e^{k^2c^2t}$$

Solution for u :

u(x, t) = X(x) T(t)

$$\mathbf{u(x, t) = (A\,e^{-kx} + B\,e^{kx})(C\,e^{k^2c^2t})}$$

1.19 Laplace Equation

Laplace equation in two dimensions is

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Laplace equation in three dimensions is

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

Solution of Laplace Equation in Two Dimensions

Laplace equation is

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \qquad \text{....(1)}$$

Let $\quad u(x, y) = X(x)\, Y(y) \qquad \ldots(2)$

where, X is a function of x only and y is a function of y only be a solution of Eq. (1).

$$\frac{\partial^2 u}{\partial x^2} = X''\,Y \text{ and } \frac{\partial^2 u}{\partial y^2} = XY''$$

Substituting in Eq. (1) we have $X''\,Y + XY'' = 0$

or $$\frac{X''}{X} = -\frac{Y''}{Y} \qquad \ldots.(3)$$

Now, LHS of Eq. (3) is a function of x only and the RHS is a function of y only. Since, x and y are independent variable, this equation can hold only when each side is equal to a constant

- When constant is negative $(-k^2)$

$$\frac{1}{X}\frac{d^2X}{dx^2} = -k^2$$

$$\therefore \quad \frac{d^2X}{dx^2} + k^2X = 0$$

$$(D^2 + k^2)\, X = 0$$

A. E. is $D^2 + k^2 = 0$

$$D^2 = -k^2$$

$$D = \pm k\, i$$

Solution is $X = A \cos kx + B \sin kx$

Also, $$-\frac{1}{y}\frac{d^2Y}{dy^2} = -k^2$$

$$\therefore \quad \frac{d^2Y}{dy^2} - k^2Y = 0$$

$$(D^2 - k^2)\, Y = 0$$

A.E. is $D^2 - k^2 = 0$

$$\therefore \quad D^2 = k^2$$

$$D = \pm k$$

Solution is $Y = Ce^{-ky} + De^{ky}$

Solution for u :

$u(x, y) = X(x)\, Y(y)$

$$\mathbf{u(x, y) = (A \cos kx + B \sin kx)\,(C\, e^{-ky} + D\, e^{ky})}$$

- When constant is zero

$$\frac{1}{X}\frac{d^2X}{dx^2}=0$$

$$\therefore \quad \frac{d^2X}{dx^2}=0$$

$$\frac{dX}{dx}=A$$

$$X = Ax + B$$

Similarly, $Y = Cy + D$

Solution for u :

$u(x, y) = X(x)\, Y(y)$

$$\mathbf{u(x, y) = (Ax + B)\,(Cy + D)}$$

- When constant is positive (k^2)

$$\frac{1}{X}\frac{d^2X}{dx^2}=k^2$$

$$\therefore \quad \frac{d^2X}{dx^2}-k^2X=0$$

$$(D^2 - k^2)\, X = 0$$

A.E. is $D^2 - k^2 = 0$

$$D^2 = -k^2$$

$$D = \pm k$$

Solution is $X = Ae^{-kx} + Be^{kx}$

Also, $\frac{-1}{Y}\frac{d^2Y}{dy^2}=k^2$

$$\therefore \quad \frac{d^2Y}{dy^2}+k^2Y=0$$

$$(D^2 + k^2)\, Y = 0$$

A.E. is $D^2 + k^2 = 0$

$$D^2 = -k^2$$

$$D = \pm ki$$

Solution for u :

$u(x, y) = X(x)\, Y(y)$

$$\mathbf{u(x, y) = (Ae^{-kx} + Be^{kx})\,(C \cos ky + D \sin ky)}$$

List of Formulae

- **Exact Differential Equation :**

  Equation : $Mdx + Ndy = 0$

  Condition : $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$

  General solution : $\displaystyle\int_{y\ \text{constant}} M \cdot dx + \int [\text{terms in N which do not involve x}]\, dy = c$

- **Integrating Factors :**
  By inspection :

Terms		I. F.	DIFFERENTIAL
xdy–ydx		$\dfrac{1}{x^2}$	$\dfrac{xdy - ydx}{x^2} = d\left(\dfrac{y}{x}\right)$
		$\dfrac{1}{xy}$	$\dfrac{dy}{y} - \dfrac{dx}{x} = d\log\left(\dfrac{y}{x}\right)$
		$\dfrac{1}{y^2}$	$\dfrac{xdy - ydx}{y^2} = d\left(\dfrac{-x}{y}\right)$
		$\dfrac{1}{x^2 + y^2}$	$\dfrac{xdy - ydx}{x^2 + y^2} = d\left(\tan^{-1}\dfrac{y}{x}\right) = -d\tan^{-1}\dfrac{x}{y}$
xdy + ydx		$\dfrac{1}{xy}$	$\dfrac{xdy + ydx}{xy} = d\log(xy)$
		$\dfrac{1}{x^2 + y^2}$	$\dfrac{xdx + ydy}{x^2 + y^2} = \dfrac{1}{2}\left[\log\left(x^2 + y^2\right)\right]$

If $Mdx + Ndy = 0$ is homogenous then

$$\text{I.F.} = \frac{1}{Mx + Ny}$$

If $Mdx + Ndy = 0$

and $M = y\, f_1(xy)$, $N = x\, f_2(xy)$ then

$$\text{I.F.} = \frac{1}{Mx - Ny}$$

If Mdx + Ndy = 0

and $\dfrac{\dfrac{\partial M}{\partial y}-\dfrac{\partial N}{\partial x}}{N}=f(x)$ function of x alone or constant

I.F. = $e^{\int f(x)dx}$

If Mdx + Ndy = 0

and $\dfrac{\dfrac{\partial N}{\partial x}-\dfrac{\partial M}{\partial y}}{M}=\phi(y)$ a function of y alone or constant.

I.F. = $e^{\int \phi(y)dy}$

- **Linear Equations :**

Equation : $\dfrac{dy}{dx}+Py=Q$

I. F. : $e^{\int Pdx}$

General solution : $y(\text{I.F.})=\int Q\,(\text{I.F.})\,dx+c$

Bernoulli's Equation : $\dfrac{dy}{dx}+Py=Qy^n$

Substitution : $t = y^{1-n}$

- **Orthogonal trajectories of family of curves**

Differential equation governing the family of orthogonal curves is

$$F\left(x,y,-\frac{1}{y'}\right)=0$$

- **Higher Order Linear Differential Equations with constant coefficients**

$$\frac{d^ny}{dx^n}+a_1\frac{d^{n-1}y}{dx^{n-1}}+a_2\frac{d^{n-2}y}{dx^{n-2}}+\ldots+a_ny=X$$

General Solution : $y = y_c + y_p$

Complementary Function

Case	Roots	Complementary Function
1.	Real and distinct	$y = C_1e^{m_1x} + C_2e^{m_2x} + ... + C_ne^{m_nx}$
2.	Real and repeated	m_1 be repeated two times m_2 be repeated three times $y = (C_1x + C_2)e^{m_1x} + (C_3x^2 + C_4x + C_5)e^{m_2x} + ...$
3.	Complex roots	$\left.\begin{array}{l} m_1 = \alpha + i\beta \\ m_2 = \alpha - i\beta \end{array}\right\}$ Conjugate Pair $y = e^{\alpha x}(C_1\cos\beta x + C_2\sin\beta x)$
4.	Repeated complex roots	$m_1 = \alpha + i\beta$ $m_2 = \alpha - i\beta$ $m_3 = \alpha + i\beta$ $m_4 = \alpha - i\beta$ $y = e^{\alpha x}\left[(C_1x + C_2)\cos\beta x + (C_3x + C_4)\sin\beta x\right] + ...$

Particular Integral:

General method :

$$\frac{1}{D-m}X = e^{mx}\int e^{-mx}X\,dx$$

$$\frac{1}{D+m}X = e^{-mx}\int e^{mx}X\,dx$$

Short method :

Type 1 :

$f(a) \neq 0 \qquad \frac{1}{f(D)}e^{ax} = \frac{1}{f(a)}e^{ax}$

$f(a) = 0 \qquad \frac{1}{f(D)}e^{ax} = \frac{1}{\phi(a)}\frac{x^re^{ax}}{r!} \qquad \phi(a) \neq 0$

Type 2a :

$f(-a^2) \neq 0 \qquad \frac{1}{f(D^2)}\sin(ax+b) = \frac{1}{f(-a^2)}\sin(ax+b)$

$f(-a^2) = 0 \qquad \frac{1}{f(D^2)}\sin(ax+b) = \frac{1}{\phi(-a^2)}\frac{(-1)^r x^r}{r!(2a)^r}\sin(ax+b+\frac{r\pi}{2})$

Type 2b :

$f(a^2) \neq 0 \qquad \frac{1}{f(D^2)}\sinh(ax+b) = \frac{1}{f(a^2)}\sinh(ax+b)$

Type 3 :

$\frac{1}{f(D)}x^m$ Expand $\left[\frac{1}{f(D)}\right]$ in ascending powers of D.

$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + ...$

Type 4 :

$\frac{1}{f(D)}e^{ax}V = e^{ax}\frac{1}{f(D+a)}V$

Type 5 :

$\frac{1}{f(D)}x\,V = \left[x - \frac{1}{f(D)}f'(D)\right]\frac{1}{f(D)}V$

$(1 + D)^{-1} = 1 - D + D^2 - D^3 + ...$

$(1 - D)^{-1} = 1 + D + D^2 + D^3 + ...$

- **Euler-Cauchy Equation**

Homogeneous Equation

$a_0x^2y'' + a_1xy' + ... + a_2y = 0$

Case	Roots	General Solution
1. Real and distinct	m_1 and m_2	$y = Ax^{m_1} + Bx^{m_2}$
2. Real and equal	m	$y = x^m (A + B\,\ell n\,x)$
3. Complex roots	$m = p \pm iq$	$y = x^p (A\cos(q\,\ell n\,x) + B\sin(q\,\ell n\,x))$

Non–Homogeneous Equation

$a_0x^2y'' + a_1xy' + a_2y = r(x) \quad a_0 \neq 0, \;\; x \neq 0$

put $x = e^z$

$x^nD^ny = \theta(\theta-1)(\theta-2).....[\theta-(n-1)]y$

- $$A\frac{\partial^2\mu}{\partial x^2}+B\frac{\partial^2\mu}{\partial x\,\partial y}+C\frac{\partial^2\mu}{\partial y^2}+f\left(x, y, \mu, \frac{\partial\mu}{\partial x}, \frac{\partial\mu}{\partial y}\right)=0$$

  The above equation is

  elliptic, if $B^2 - 4AC < 0$

  hyperbolic, if $B^2 - 4AC > 0$

  parabolic, if $B^2 - 4AC = 0$

- **Wave equation**

  $$\frac{\partial^2 y}{\partial t^2}=c^2\frac{\partial^2 y}{\partial x^2}$$

  $y(x, t) = X(x)\,T(t)$

  $$\frac{1}{T}\cdot\frac{d^2T}{dt^2}=\frac{c^2}{X}\cdot\frac{d^2X}{dx^2} = \text{constant}$$

  When constant is negative $(-k^2)$

  $$y(x, t) = \left(A\cos\frac{kx}{c}+B\sin\frac{kx}{c}\right)\left(C\cos kt+D\sin kt\right)$$

  When constant is zero

  $y(x, t) = (Ax + B)\,(Ct + D)$

  When constant is positive (k^2)

  $$y(x, t) = \left(A\,e^{-\frac{kx}{c}} + B\,e^{\frac{kx}{c}}\right)\left(C\,e^{-kt}+De^{-kt}\right)$$

- **Heat equation**

  $$\frac{\partial u}{\partial t} = c^2\frac{\partial^2 u}{\partial x^2}$$

  $u(x, t) = X(x)\,T(t)$

  $$\frac{X''}{X}=\frac{1}{c^2}\cdot\frac{T'}{T} = \text{constant}$$

  When constant is negative $(-k^2)$

  $$u(x, t) = \left(A\cos kx+B\sin kx\right)\left(C\,e^{-k^2c^2t}\right)$$

When constant is zero

$u(x, t) = Dx + E$

When constant is positive (k^2)

$u(x, t) = (A e^{-kx} + B e^{kx})(C e^{k^2c^2t})$

- **Laplace equation**

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$u(x, y) = X(x)\, Y(y)$

$$\frac{X''}{X} = -\frac{Y''}{Y} = \text{constant}$$

When constant is negative ($-k^2$)

$u(x, y) = (A \cos kx + B \sin kx)\,(C e^{-ky} + D e^{ky})$

When constant is zero

$u(x, y) = (Ax + B)\,(Cy + D)$

When constant is positive (k^2)

$u(x, y) = (Ae^{-kx} + Be^{kx})\,(C \cos ky + D \sin ky)$

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Assignment – 1

Duration : 45 Min. **Max. Marks : 30**

Q 1 to Q 6 carry one mark each

1. The order and degree of the differential equation

$$\frac{d}{dx}\left\{x^2\left(\frac{d^2y}{dx^2}\right)^3\right\}+\cos(xy)=e^x \text{ is}$$

(A) order = 2, degree = 1

(B) order = 3, degree = 1

(C) order = 1, degree = 3

(D) order = 2, degree = 2

2. The type of partial differential equation

$$\frac{\partial^2 P}{\partial x^2}+\frac{\partial^2 P}{\partial y^2}+3\frac{\partial^2 P}{\partial x\partial y}+2\frac{\partial P}{\partial x}-\frac{\partial P}{\partial y}=0 \text{ is}$$

[CE – 2016]

(A) elliptic (B) parabolic

(C) hyperbolic (D) none of these

3. The degree of the differential equation

$$\left[y+x\left(\frac{d^2y}{dx^2}\right)^3\right]^{\frac{1}{4}}=\frac{d^3y}{dx^3}$$

(A) 4 (B) 3

(C) 2 (D) 1

4. If y is the solution of the differential equation $y^3\frac{dy}{dx}+x^3=0,\ y(0)=1,$ the value of y (–1) is **[ME – 2018]**

(A) –2 (B) –1

(C) 0 (D) 1

5. The general solution of the differential equation $\frac{dy}{dx}=\frac{1+\cos 2y}{1-\cos 2x}$ is

[EC – 2015]

(A) tan y – cot x = c (c is a constant)

(B) tan x – cot y = c (c is a constant)

(C) tan y + cot x = c (c is a constant)

(D) tan x + cot y = c (c is a constant)

6. Solution of

$(2x^2y + 4x^3 - 12\,xy^2 + 3y^2 - x\,e^y + e^{2x})\,dy$
$+ (12x^2y + 2xy^2 + 4x^3 - 4y^3 + 2ye^{2x} - e^y)\,dx$
$= 0$ is

(A) $4x^3y + x^2y^2 + x^4 + C$

(B) $4x^3y + x^4 - 4y^3x + ye^{2x} + C$

(C) $4x^3y + x^2y^2 + x^4 - 4y^3x + ye^{2x} - x\,e^y + y^3$

(D) $-4y^3x + ye^{2z} - e^yx + C$

Q7 to Q18 carry two marks each

7. P. I. of $\frac{d^6y}{dx^6}+y=\cos x$ is

(A) $\frac{1}{3}\frac{x\sin x}{2}$ (B) $\frac{1}{4}x\sin x$

(C) $\frac{1}{8}x\sin x$ (D) $\frac{1}{5}\frac{x\sin x}{2}$

8. Particular integral of

$$\frac{d^2y}{dx^2}+\frac{dy}{dx}+y=\sin 2x \text{ is}$$

(A) $\frac{1}{11}$ [2 cos 2x + 3 sin 2x]

(B) $\frac{1}{3}$ [2 sin 2x + 3 sin x]

(C) $-\frac{1}{11}$ [2 cos x + 3 sin x]

(D) $-\frac{1}{13}$ [2 cos 2x + 3 sin 2x]

9. Particular integral of

$$\frac{d^2y}{dx^2}+2\frac{dy}{dx}+2y=x$$

(A) x–1 (B) $\frac{1}{2}$ (x–1)

(C) 1–x (D) $-\frac{1}{2}$ (x–1)

10. Evaluate P. I. of $(D^4 + D^2)y = 96x^2$

(A) $8x^4 - 96x^2 + 192$

(B) $6x^2 - 96x + 1$

(C) $8x^4 + 96x^2 - 192$

(D) $6x^2 + 96x - 1$

11. P. I. of $\left(\frac{d^2y}{dx^2}+3\frac{dy}{dx}+2y\right)=x^2$ is

(A) $\frac{1}{2}(x^2 - 3x + \frac{7}{2})$

(B) $\frac{1}{2}(3x + 7)$

(C) $\frac{2}{3}(x^3 - 4x + \frac{4}{3})$

(D) none of these

12. P. I. of $\frac{d^2y}{dx^2}-4\frac{dy}{dx}+3y=e^x\cos 2x$ is

(A) $\frac{e^x}{8}$ (cos 2x + cot 2x)

(B) $-\frac{e^x}{8}$ (sin 2x + cos 2x)

(C) $\frac{e^x}{8}$ (sin 2x + cos 2x)

(D) e^x(sin 2x – cos2x)

13. Particular solution of

$$\frac{d^3y}{dx^3}-\frac{3d^2y}{dx^2}+3\frac{dy}{dx}-y=x^2e^x \text{ is}$$

(A) $\frac{x^5}{60}$ (B) $\frac{e^x x^4}{20}$

(C) $\frac{e^x x^5}{60}$ (D) $\frac{e^x}{60}$

14. Consider the differential equation

$$\frac{d^2x(t)}{dt^2}+3\frac{dx(t)}{dt}+2x(t)=0.$$

Given x(0) = 20 and x(1) = 10/e, where e = 2.718, the value of x(2) is _____.

[EC – 2015]

15. Consider the ordinary differential equation $x^2\frac{d^2y}{dx^2}-2x\frac{dy}{dx}+2y=0$. Given the values of y(1) = 0 and y(2) = 2, the value of y(3) (round off to 1 decimal place), is _____. **[CE – 2019]**

16. The general solution of the differential equation $\frac{dy}{dx} = \cos(x + y)$, with c as a constant, is **[ME – 2014]**

(A) $y + \sin(x + y) = x + c$

(B) $\tan\left(\frac{x+y}{2}\right) = y + c$

(C) $\cos\left(\frac{x+y}{2}\right) = x + c$

(D) $\tan\left(\frac{x+y}{2}\right) = x + c$

17. Find the solution of $\frac{d^2y}{dx^2} = y$ which passes through the origin and the point $\left(\ell n2, \frac{3}{4}\right)$. **[ME – 2015]**

(A) $y = \frac{1}{2}e^{x} - e^{-x}$

(B) $y = \frac{1}{2}(e^{x} + e^{-x})$

(C) $y = \frac{1}{2}(e^{x} - e^{-x})$

(D) $y = \frac{1}{2}e^{x} + e^{-x}$

18. For the Ordinary Differential Equation $\frac{d^2x}{dt^2} - 5\frac{dx}{dt} + 6x = 0$, with initial conditions $x(0) = 0$ and $\frac{dx}{dt}(0) = 10$, the solution is **[CE–2020]**

(A) $-5e^{2t} + 6e^{3t}$

(B) $5e^{2t} + 6e^{3t}$

(C) $-10e^{2t} + 10e^{3t}$

(D) $10e^{2t} + 10e^{3t}$

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Test Paper – 1

Duration : 30 Min. **Max. Marks : 25**

Q1 to Q5 carry one mark each

1. If the general solution of L.D.E. with constant coefficients f(D) = 0 be $c_1e^{a_1x} + c_2e^{a_2x} + ... + c_ne^{a_nx} = 0$, then roots of auxiliary equation are

(A) all real

(B) all imaginary

(C) some real and some imaginary

(D) all real and distinct

2. The auxiliary equation $aD^2 + bD + c = 0$ can be obtained from L.D.E. where a, b, c are constant by substituting

(A) $y = x^m$ (B) $y = b \sin mx$

(C) $y = m^x$ (D) $y = e^{nx}$

3. If f(D)y = 0 be a linear differential equation with constant coefficients then its auxiliary equation is

(A) $f(m^3) = 0$ (B) f(m) = 0

(C) f(D – m) = 0 (D) f(x) = 0

4. The general solution of L.D.E. with constant coefficients f(D)y = x is equal to the sum of

(A) particular solution of f(D)y = x and general solution of f(D)y = 0

(B) particular solution of f(D)y = x and general solution of f(D – a) = 0

(C) particular solution of f(D)y = 0 and general solution of f(D)y = x

(D) all of these

5. If the Differential equation is of type f(D)y = sin ax, where f(D) is a polynomial containing odd powers in D only then

(A) no particular integral can always be obtained

(B) a particular integral can always be obtained

(C) P.I. in this case is taken as constant

(D) P.I. is $\frac{\sin ax}{f(-a^2)}$ if $f(-a^2) \neq 0$

Q6 to Q15 carry two marks each

6. The general solution of L.D.E.$(D^4 - 81)$ y = 0 is given by

(A) $(c_1 + c_2x)e^{3x} + (c_3 + c_4x)\sin 3x$

(B) $(c_1 + c_2)e^{3x} + (c_3 + c_4x)e^{-3x}$
$+ (c_5 + c_6x)\cos 3x + (c_7 + c_8x)\sin 3x$

(C) $c_1e^{3x} + c_2e^{-3x} + c_3\cos 3x + c_4\sin 3x$

(D) $c_1e^{3x} + c_2e^{-3x}$
$+ e^{3x}(c_3\cos x + c_4\sin x)$

7. $\frac{1}{D-a}Q(x)$ is equal to

(A) $e^{ax}\int Q(x)dx$

(B) $e^{-ax}\int e^{ax}Q(x)dx$

(C) $e^{-ax}\int Q(x)dx$

(D) $e^{ax}\int e^{-ax}Q(x)dx$

8. The general solution of D.E.

$D^2(D+1)^2 y = e^x$ is

(A) $y = c_1 + c_2x + (c_3 + c_4x)e^x$

(B) $y = c_1 + c_2x + (c_3 + c_4x)e^{-x} + \frac{1}{4}e^x$

(C) $y = (c_1 + c_2e^{-x}) + (c_3 + c_4)e^{-x} + \frac{1}{4}e^x$

(D) None of these

9. The P.I. of $(D^2 - 2D)y = e^x \sin x$ is

(A) $-\frac{1}{2}e^x \sin x$ (B) $e^x \cos x$

(C) $-\frac{1}{2}e^x \cos x$ (D) None of these

10. P.I. of D.E. $(D + 2)(D - 1)^3 y = e^x$ is

(A) $\frac{1}{18}x^3e^x$ (B) x^3e^x

(C) $\frac{1}{3}x^3e^x$ (D) None of these

11. The general solution of D.E. $(D^2 - m^2)$ $y = 0$ is

(A) $y = (c_1 + c_2x)e^{mx}$

(B) $y = c_1 \sin x + c_2 \cos x$

(C) $y = c_1 \cos hx + c_2 \sin hx$

(D) None of these

12. The solution of D.E. $\frac{d^2y}{dx^2} + y = \cos 2x$ is

(A) $A \cos x + B$ is in $x + \frac{1}{3}\cos 2x$

(B) $A\cos x + B\sin x + \frac{1}{3}\sin 2x$

(C) $A\cos x + B\sin x - \frac{1}{3}\cos 2x$

(D) $A\cos x + B\sin x - \frac{1}{3}\sin 2x$

13. The P.I. of D.E. $(D^3 - D)\, y = e^x + e^{-x}$ is

(A) $\frac{1}{2}(e^x + e^{-x})$ (B) $\frac{1}{2}x(e^x + e^{-x})$

(C) $\frac{1}{2}x^2(e^x + e^{-x})$ (D) $\frac{1}{2}x^2(e^x - e^{-x})$

14. The general solution of D.E.

$\frac{d^2y}{dx^2} + n^2y = 0$ is

(A) $c_1\sqrt{\cos nx} + c_2\sqrt{\sin nx}$

(B) $c_1 \cos nx + c_2 \sin nx$

(C) $c_1 \cos^2 nx + c_2 \sin^2 nx$

(D) $c_1 \cos^3 nx + c_2 \sin^3 nx$

15. The solution of D.E. $\frac{d^2y}{dx^2} = x$ is

(A) $y = \frac{x^3}{6} + c_1x^2 + c_2$

(B) $y = \frac{x^3}{6} + c_1x + c_2$

(C) $y = \frac{x^2}{6} + c_1x + c_2$

(D) $y = -\frac{x^2}{6} + c_1x + c_2$

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